

NEWPORT 2835-C MANUAL



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Model 2835-C

Multi-Function Optical Meter

Model 2835-C

Multi-Function Optical Meter

Warranty

Newport Corporation warrants this product to be free from defects in material and workmanship for a period of 1 year from the date of shipment. If found to be defective during the warranty period, the product will either be repaired or replaced at Newport's option.

To exercise this warranty, write or call your local Newport representative, or contact Newport headquarters in Irvine, California. You will be given prompt assistance and return instructions. Send the instrument, transportation prepaid, to the indicated service facility. Repairs will be made and the instrument returned, transportation prepaid. Repaired products are warranted for the balance of the original warranty period, or at least 90 days.

Limitation of Warranty

This warranty does not apply to defects resulting from modification or misuse of any product or part. This warranty also does not apply to fuses, batteries, or damage from battery leakage.

This warranty is in lieu of all other warranties, expressed or implied, including any implied warranty of merchantability or fitness for a particular use. Newport Corporation shall not be liable for any indirect, special, or consequential damages.

Statement of Calibration

This instrument has been inspected and tested in accordance with specifications published by Newport Corporation.

The accuracy and calibration of this instrument and photodetector (where applicable) is traceable to the National Institute for Standards and Technology through equipment which is calibrated at planned intervals by comparison to the certified standards maintained at Newport Corporation.



EC DECLARATION OF CONFORMITY

Model 2835-C

We declare that the accompanying product, identified with the "CE" mark, meets the intent of the Electromagnetic Compatibility Directive, 89/336/EEC and Low Voltage Directive 73/23/EEC.

Compliance was demonstrated to the following specifications:

EN50081-1 EMISSIONS:

Radiated and conducted emissions per EN55011, Group 1, Class A

EN50082-1 IMMUNITY:

Electrostatic Discharge per IEC 1000-4-2, severity level 3
Radiated Emission Immunity per IEC 1000-4-3, severity level 2
Fast Burst Transients per IEC 1000-4-4, severity level 3
Surge Immunity per IEC 1000 4-5, severity level 3

IEC SAFETY:

Safety requirements for electrical equipment specified in IEC 1010-1.

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Safety Symbols and Terms

The following safety terms are used in this manual:

The **WARNING** heading in this manual explains dangers that could result in personal injury or death.

The **CAUTION** heading in this manual explains hazards that could damage the instrument.

In addition, a **NOTES** heading gives information to the user that may be beneficial in the use of this instrument.

GENERAL WARNINGS AND CAUTIONS

The following general warnings and cautions are applicable to this instrument:

WARNING

This instrument is intended for use by qualified personnel who recognize shock hazards or laser hazards and are familiar with safety precautions required to avoid possible injury. Read the instruction manual thoroughly before using, to become familiar with the instrument's operations and capabilities.

WARNING

The American National Safety Institute (ANSI) states that a shock hazard exists when probes or sensors are exposed to voltage levels greater than 42VDC or 42V peak AC. Do not exceed 42V between any portion of the Model 2835-C (or any attached detector or probe) and earth ground or a shock hazard will result.

CAUTION

There are no user serviceable parts inside the Model 2835-C. Work performed by persons not authorized by Newport may void the warranty. For instructions on obtaining warranty repair or service please refer to Section 5 of this manual.

Definitions

A	amps
AC	alternating current
ADC	analog-to-digital converter
BAT	battery option
BIC	biconic fiber connector
BNC	standard coaxial connector type
C	degrees Centigrade
DC	direct current
F	degrees Fahrenheit
fA	femtoamps
Hz	hertz (cycles per second)
I-V	current-to-voltage converter
kHz	kilohertz
LSD	least significant digit
k	kiloOhms
mA	milliamps
mV	millivolts
nA	nanoamps
nF	nanofarads
nm	nanometers
pA	picoamps
P-P	peak-to-peak
RH	relative humidity
S/N	serial number
μA	microamps
μs	microsecond
V	volts
W	watts

Specifications

Physical Specifications:

Dimensions:	4.2 x 8.8 x 13.9 in (107 x 224 x 353 mm)
Weight:	8 lb, 3 oz (3.7 kg)
Enclosure:	Metal case, painted
Connectors:	(2) 8-Pin Sub Mini DIN CAL MODULE Inputs; (2) BNC Analog Outputs, Trigger Output and Trigger Input; 9 Pin D-Sub RS-232, 24 Conductor GPIB
Power:	100/120/220/240 VAC $\pm$ 10%, 50/60 Hz
Display:	5.5 digit annunciated VFD
Display Update Rate:	10 Hz
Gain Ranges:	Up to 7 decades (Detector and MODE dependent)
Operating Environment:	0°C - 40°C; < 70% RH noncondensing
Storage Environment:	-20°C - 60°C; < 90% RH noncondensing
Compatible Detectors:	Low-Power (Semiconductor) Family High-Power (Thermopile) Family Energy (Pyroelectric) Family

Electrical Specifications:

DC Current Measurement (Low-Power, Semiconductor Photodiode CAL MODULE)

Signal Range: ^{1,2}	0	1	2	3	4	5	6
Full-Scale Current: ⁴	2.51 nA	25.1 nA	251 nA	2.51 μ A	25.1 μ A	251 μ A	2.50 mA
A/D Resolution: (20,000 Count Precision)	126 fA	1.26 pA	12.6 pA	126 pA	1.26 nA	12.6 nA	126 nA
Display Noise Floor: (Input Open, Filter Off)	≤ 8 LSD	≤ 1 LSD	≤ 1 LSD	≤ 1 LSD	≤ 1 LSD	≤ 1 LSD	≤ 1 LSD
Full-Scale Accuracy: ³ (Typical)	$\pm 0.1\%$	0.05%	0.05%	0.05%	0.05%	0.05%	0.05%
Full-Scale Accuracy: ³ (Worst-Case)	$\pm 0.48\%$	.30%	.30%	.30%	.30%	.30%	.30%

Peak-Peak Current Measurement (Low-Power, Semiconductor Photodiode CAL MODULE)

Signal Range: ^{1, 2}	0	1	2	3	4	5	6	7	8
Full Scale (P-P) Current: ⁴	253 nA	797 nA	2.52 µA	7.97 µA	25.1 µA	79.3 µA	251 µA	793 µA	2.51 mA
A/D Resolution:	61.7 pA	195 pA	616 pA	1.95 nA	6.13 nA	19.4 nA	61.3 nA	194 nA	613 nA
Full-Scale Accuracy: ³ (Typical)	±1%	±1%	±1%	±1%	±1%	±1%	±1%	±1%	±1%
Full-Scale Accuracy: ³ (Worst Case)	±2%	±2%	±2%	±2%	±2%	±2%	±2%	±2%	±2%
Bandwidth (3db):	(5 Hz - 1 kHz)			(5 Hz - 10 kHz)			(5 Hz - 47 kHz)		
Frequency Range for ±2% (Typ) Accuracy:	(50 Hz - 100 Hz)			(50 Hz - 1.4 kHz)			(50 Hz - 7 kHz)		
Trigger Level: 4% of full scale (fixed) above ground.									

¹ Listed signal ranges specify meter capability. Available signal ranges are detector dependent.

² Maximum measurable signal is detector dependent. See description of detector saturation message "SA", page 83.

³ After 60 min warm-up, followed by execution of AUTOCAL command. See Section 2.4.4.

⁴ Full scale current may vary due to AUTOCAL compensation of DC offsets.

Analog Output

Full Scale Voltage: 0 - 2.5V into 50 Ω

Accuracy: $\pm 2.5\%$

D.C. Voltage Measurement (Thermopile CAL MODULE)

Signal Range: ^{1,2}	0	1	2	3
Full Scale Voltage: ⁴ (20,000 Count Precision)	2.49 mV	24.9 mV	249 mV	2.49 V
Resolution: (20,000 Count Precision)	125 nV	1.25 μ V	12.5 μ V	125 μ V
Display Noise Floor: (Input Shorted)	≤ 8 LSD	≤ 1 LSD	≤ 1 LSD	≤ 1 LSD
Full Scale Accuracy (Typ.): ³	$\pm .3\%$	$\pm .2\%$	$\pm .1\%$	$\pm .05\%$
Full Scale Accuracy: (Worst Case)	$\pm .56\%$	$\pm .36\%$	$\pm .18\%$	$\pm .1\%$
Bandwidth (3db):	5 Hz	5 Hz	5 Hz	5 Hz

Pulse Voltage Measurement (Energy, Pyroelectric CAL MODULE)

Signal Range: ¹	0	1	2	3	4	5	6	7	8	9
Full Scale Voltage: ^{2,4}	789 μ V	2.50 mV	7.89 mV	25.0 mV	78.9 mV	250 mV	789 mV	2.50 V	7.91 V	25.0 V
A/D Resolution:	193 nV	610 nV	1.93 μ V	6.10 μ V	19.3 μ V	61.0 μ V	193 μ V	610 μ V	1.93 mV	6.10 mV
Full-Scale Accuracy: ³ (Typical)	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$	$\pm 1\%$
Full-Scale Accuracy: ³ (Worst Case)	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$	$\pm 2\%$

Maximum Pulse Repetition Rate: 2 kHz

Trigger Level: 8% of Full Scale¹ (fixed)

¹ Listed signal ranges specify meter capability. Available signal ranges are detector dependent.

² Full scale voltage is measured relative to baseline voltage.

³ After 60 min warm-up, followed by AUTOCAL command. See Section 2.4.4.

⁴ Full scale voltage may vary due to AUTOCAL compensation of DC offsets.

Analog Output

Full Scale Voltage: 0 - 2.5V into 50 Ω

Accuracy: $\pm 5.0\%$

Detector Signals and Calculations:

S	Represents the most recent signal value obtained from the A/D converter. It may represent amps or volts and may be analog and or digitally filtered.
S_d	Represents the value stored as a reference signal for subsequent use in signal offset, i.e. ZERO calculations. $S_d=0$ when ZERO is off.
$S - S_d$	Represents the most recent net signal value. This is the value that is displayed when units are set to Amps or Volts. Note that $S_d = 0$ when ZERO is off.
$\frac{(S - S_d)}{R_\lambda}$	Measurement calculation when the display units are Watts or Joules. R_λ is the detector responsivity associated with the current wavelength setting.
$10\log\left(\frac{(S - S_d)/R_\lambda}{1\text{mW}}\right)$	Measurement calculation when the display units are ten times the (base ten) logarithm of the ratio of the measured power to 1 mW, i.e. dBm.
$\frac{(S - S_d)/R_\lambda}{\text{STO-REF}}$	Measurement calculation when the display units are the ratio of measured power to the value stored by the STO-REF function.
$10\log\left(\frac{(S - S_d)/R_\lambda}{\text{STO-REF}}\right)$	Measurement calculation when the display units are ten times the (base ten) logarithm of the ratio of measured power to the value stored by the STO-REF function.

Section 1

General Information

1.1 System Review

The Model 2835-C Multi-Function Optical Meter is a high performance instrument with a wealth of measurement and triggering features designed to provide measurement sensitivity, flexibility and speed. In spite of its power, Model 2835-C is also designed to provide simple operation with direct panel access to basic features and a shallow menu for access to advanced features.

Great flexibility exists within the command structure of the Model 2835-C so that even complex measurements can be set up quickly and easily. The 2835-C can react to or provide triggering, act as an exposure or noise meter, or data log up to 1,000 measurements per channel!

The Model 2835-C is compatible with all of Newport's Low-Power, High-Power and Energy detector families. A family tree of the 2835-C compatible detectors and accessories is shown in Figure 1 below.

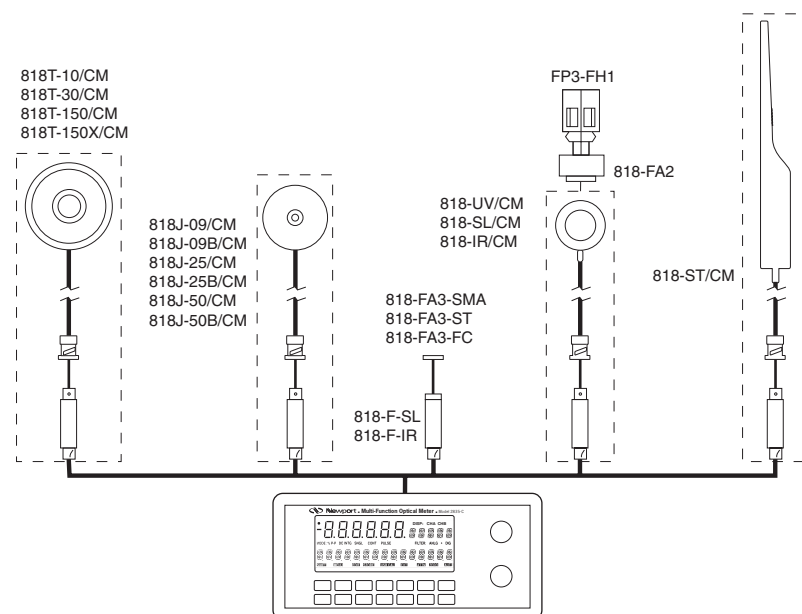


Figure 1 – Model 2835-C Controller and Accessories

The Model 2835-C connects to detectors through a calibration module containing information unique to the detector being used. Calibration modules are ordered with a detector at the time of purchase and are labeled with the detector's model number and serial number. Detectors with calibration modules have a "/CM" appended to their model number.

EXAMPLE: 818-SL (no calib. module) 818-SL/CM (with calibration module)

1.2 Scope of This Manual

Please carefully read this instruction manual before using the Model 2835-C Multi-Function Optical Meter. Be especially careful to observe the warnings and cautions throughout this manual. If any operating instructions are not clear, contact Newport Corporation. This instruction manual contains the necessary information for operation and maintenance of the Newport Model 2835-C Multi-Function Optical Meter as well as information for troubleshooting and obtaining service if necessary. This information is divided into the following sections:

Section 1	General Information and Functional Description
Section 2	System Operation
Section 3	Principles of Operation
Section 4	Computer Interfacing
Section 5	Remote Command Reference
Section 6	Maintenance, Test, and Troubleshooting
Section 7	Factory Service
Appendix A	Syntax and Definitions
Appendix B	Error Messages
Appendix C	Status Reporting System
Appendix D	Sample Programs

1.3 Unpacking and Inspection

All Model 2835-C Multi-Function Optical Meters are carefully assembled, tested and inspected before shipment. Upon receiving this instrument, check for any obvious signs of physical damage that might have occurred during shipment. Report any such damage to the shipping agent immediately. Retain the original packing materials in case reshipment becomes necessary.

1.4 Preparation for Use

The Model 2835-C Multi-Function Optical Meter should have some operations performed before measurements are made. These include:

Connecting AC Power (Section 2.5)

Detector Connection and Setup (Section 2.6)

1.5 Optional Accessories and Services

The Newport Catalog presents up-to-date information on detectors, detector accessories and detector calibration services available for use with the Model 2835-C. Refer to Figure 1 for the Model 2835-C family tree of accessories.

Section 2

System Operation

2.1 Introduction

The Model 2835-C is designed to provide quick operation and to avoid a steep learning curve. This section starts by giving a brief listing of display, key pad and menu command features. Much of the Model 2835-C's operation will be obvious after these descriptions. The manual then explains each key operation and menu command in detail. Reference each of these detailed explanations as required when getting started.

2.2 Display

The Model 2835-C incorporates a vacuum fluorescent display, VFD, which can be clearly observed with most laser goggles and at high angles of incidence. Figure 2 illustrates and identifies the primary regions and annunciators within the Model 2835-C's display.

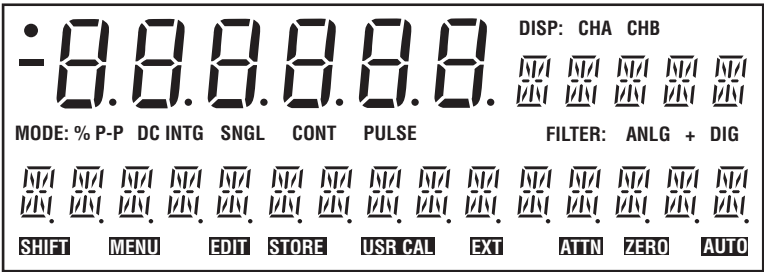


Figure 2a – Model 2835-C VFD Display

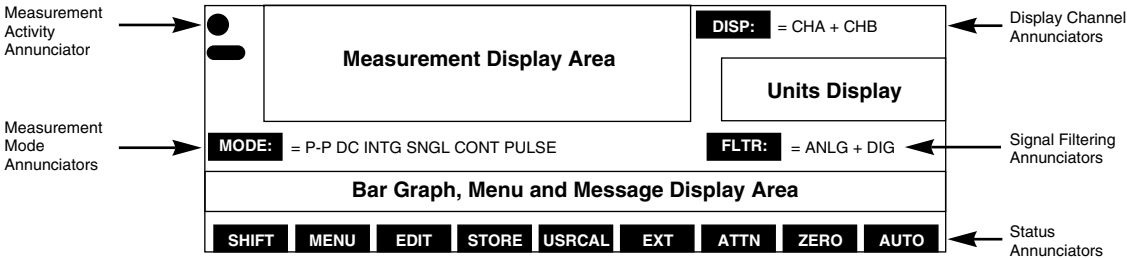


Figure 2b – Description of Model 2835-C Display Regions

Table 1 – Model 2835-C Display Annunciators

Annunciator	Comment
CH A	The display is showing Channel A information.
CH B	The display is showing Channel B information.
●	Blinking indicates that the meter is making measurements.
DC	Meter is set to make DC signal measurements.
INTG	Meter is set to make INTEGRATED signal measurements.
P-P	Meter is set to make Peak-to-Peak signal measurements.
SNGL	Meter will make only one measurement per front panel or external trigger.
CONT	Meter will continuously make measurements until stopped.
PULSE	Meter is set to make pulse energy measurements.
ANLG	Analog signal filtering is on.
DIG	Digital signal filtering is on.
SHIFT	Indicates that the next key press will execute a blue key function.
MENU	Meter and display are in menu mode. Measurement is stopped.
STORE	Meter is logging measurements into memory, i.e. data logging.
USRCAL	User (versus calibration module) has supplied the detector responsivity in use.
EXT	External trigger input is enabled.
AUTO	Automatic signal gain ranging is on.
ZERO	Background signal subtraction (zeroing) is on.
ATTN	The responsivity in use includes the affect of the detector's attenuator.

2.3 Top Level Key Functions

Operating controls for the Model 2835-C are found on both the front and rear panels of the Model 2835-C. For measurement operation, only the front panel controls are used. Rear panel controls are used for AC line power setup, Section 2.5. The front panel key pad of the Model 2835-C, Figure 3, provides quick access to measurement functions and menu access to advanced features and setup parameters. Most of the keys on the front panel only affect

the channel indicated by the display channel annunciators. Table 2 and Sections 2.3.1 through 2.3.21 list and describe each top level key function. Menu level key functions are discussed in Section 2.4.

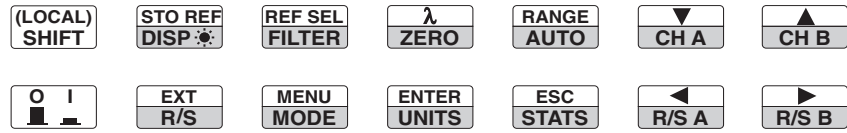


Figure 3 – Front Panel Key Pad

Table 2 – Top Level Key Functions and Associated Remote Commands

Keypad	Remote Commands	Description
 O I	None	Turns the Model 2835-C on and off.
SHIFT	None	Enables the blue key functions.
DISP 	DISP, DISP?	Cycles display brightness: OFF, LOW, NORM and HIGH.
FILTER	FILTER_n, FILTER_n?	Cycles signal filtering: OFF, ANLG, DIG and ANLG+DIG.
ZERO	ZERO_n, ZERO_n?, STOZERO_n, ZEROVAL_n?	Zeroes the display via offset subtraction.
AUTO	AUTO_n, AUTO_n?	Turns automatic gain ranging on and off.
CH A	DISPCH, DISPCH?	Selects channel A as the display channel.
CH B	DISPCH, DISPCH?	Selects channel B as the display channel.
STO REF	STOREF_n, STOREF_n?	Stores last reading for future dB and REL measurements.
REF SEL	REFSEL_n, REFSEL_n?	Selects the reference source to be used for dB and REL measurements.
λ	LAMBDA_n, LAMBDA_n?	Displays and edits the calibration wavelength in use.
RANGE	RANGE_n, RANGE_n?	Displays signal gain range in use.
 , 	None	Adjusts gain range or parameter values or moves through a list.
R/S	RUN, STOP	Toggles the signal acquisition of both channels.
MODE	MODE_n, MODE_n?	Cycles meter between allowed measurement modes.
UNITS	UNITS_n, UNITS_n?	Cycles display measurement units between those allowed.
STATS	STSIZE_n, STSIZE_n?, STMAX_n?, STMIN_n?, STMXMN_n?, STMEAN_n, STSDEV_n?	Displays statistics.: Max, Min, Max-Min, Mean and Std. Dev.
R/S A	RUN_A, RUN_A?	Starts and stops channel A signal acquisition.

R/S B	RUN_B, RUN_B?	Starts and stops channel B signal acquisition.
EXT	EXT	Enable or disable external triggering.
MENU	None	Enter or exit MENU command mode.
ENTER	None	Select next lower menu level or enter edit mode.
ESC	None	Escape to next higher menu level or escape edit mode.
◀, ▶	None	Zooms bargraph in or out. When in Edit mode, selects the digit to be edited by the ▼, ▲ keys.

Note: n in a remote command stands for A (channel A) or B (channel B).

2.3.1 , Power

The  key toggles the Model 2835-C on and off. To turn the meter on, depress the  key in until it clicks and stays in its depressed position. To turn the meter off, press the key again until it clicks and rebounds to its original length.

2.3.2 SHIFT

When SHIFT is pressed, the SHIFT display annunciator lights and the blue key functions (such as STOREF) are enabled. The next key press will cause that blue function to be executed and the “shift status” to disable.

2.3.3 DISP , Display Brightness

This key cycles the display and the backlit key pad through: OFF, LOW, NORM and HIGH brightness levels. This allows a user to operate in a dark environment without light pollution from the display, (except for one dim scanning decimal point and the measurement activity annunciator). When the display is OFF, the R/S, R/S A, or R/S B keys may be used to manually trigger measurements. Any other key press returns the display to the LOW brightness state while ignoring the key function.

2.3.4 FILTER, Signal Filtering

Press this key to cycle input signal filtering between: OFF, ANLG, DIG, ANLG+DIG. This function provides methods of lowering the noise observed in the measurement data and the analog output. When the ANLG annunciator is lit, a 5Hz low pass filter lowers the noise floor by attenuating high frequency signal components. ANLG filtering is not available to High Power detectors, Energy detectors or any P-P modes.

When the DIG display annunciator is lit, measurements pass through a moving 10-sample averaging buffer before being further processed, stored or communicated to the display or computer interfaces. With DIG on, all observable values represent digitally averaged results relative to the original A/D conversions. This averaging is independent of the subsequent processing available through the STATS buffer, Section 2.3.15.

2.3.5 ZERO, Offset Subtraction

This key turns offset subtraction on and off. When turned on, the ZERO annunciator lights and the last signal reading is saved as S_d and subtracted from all subsequent signal readings S . This causes subsequent signal calculations (and the display) to use the value $S - S_d$ instead of S .

Offset subtraction allows one to remove the effects of ambient DC signals, by zeroing the display before making a measurement. Pressing the ZERO key a second time turns off the ZERO annunciator and stops offset subtraction.

2.3.6 AUTO, Automatic Gain Ranging

The AUTO key toggles automatic signal ranging on and off. When on, the AUTO annunciator lights and the signal range (amplifier gain) is adjusted to utilize maximum analog-to-digital converter resolution. When AUTO is turned off, the AUTO annunciator is turned off and the signal range is left in its current state.

Signal range can be manually controlled by the $\blacktriangledown$, $\blacktriangle$ arrow keys. Pressing an $\blacktriangledown$, $\blacktriangle$ arrow key when AUTO is on (and STATS, MENU and λ are off), turns AUTO off and executes the manual range change. See RANGE, Section 2.3.11.

Signal range changes will often not coincide with observable changes to the display value as the display is scaled by the detector responsivity and so must adjust independently. Signal range gains will occur in 1 decade steps when auto ranging and 1 or $1/2$ decade steps (depending on MODE) when manual ranging.

When auto ranging in CONT PULSE mode, the arrival rate of pulses must be above 1 Hz. In P-P CONT mode, the arrival rate of peaks must be above 50 Hz.

AUTO gain ranging is not allowed in SNGL measurement modes. When exiting a SNGL measurement MODE, AUTO will turn back on if it was on when the SNGL measurement mode was entered. (See MODE, Section 2.3.13)

2.3.7 CH A (B), Display Channel A (or B)

CH A causes channel A to be the display channel. CH B causes channel B to be the display channel. The display channel annunciators will indicate which channel is currently the display channel. When a display channel is selected, the front panel will change to reflect its settings and measurement data. Operations from the keypad or menu that affect a specific channel will only affect the currently displayed channel and not the other.

2.3.8 STOREF, Store Reference Value

STOREF causes the last measurement, D , to be stored as D_{ref} for subsequent use in relative measurement calculations. When units are relative, REL, and the stored reference is selected, the displayed value is the ratio D/D_{ref} . When units are logarithmic relative, dB, and the stored reference is selected the displayed value is the function $10 \log (D/D_{ref})$. D_{ref} is always a power reading stored in the units of Watts. Press STOREF to cause a new D_{ref} to overwrite the existing D_{ref} value.

NOTE

When not using remote interface operation and when displaying relative dB or REL measurements with the stored reference selected, the message display area will show the STOREF value used in the calculation.

2.3.9 REFSEL, Select Reference

The REF SEL key is used to select the reference value, Dref, to be used for relative, REL, and logarithmic relative, dB, unit readings. Press the REF SEL key to display the reference source in the message display area. The EDIT annunciator will be lit and the current selection will blink. Press ▼, ▲ to select a new reference source. The choices will be:

STO REF - The last stored measurement

USR REF - User defined reference (see the USRREF command in Section 5.3 Device Dependent Commands)

CH A(B) - The other channel

Press ENTER to accept the new reference source and exit the select reference mode. Press REF SEL or ESC to exit without changing the reference source.

NOTE

The CH A(B) source is only applicable if the other channel is available. If the other channel is not available then N/A will appear next to CH A(B) and you will not be able to select it.

2.3.10 λ , Wavelength

A detector calibration module contains responsivity data for its assigned detector at discrete wavelengths. By telling the meter which wavelength is being measured, the correct responsivity value is used in calculating the measured power or energy. When a wavelength falls between two calibration points, linear interpolation is used to approximate the true responsivity value.

Press the λ key to display the measurement wavelength in the message display area. Press ENTER to light the EDIT annunciator and cause the last digit of the wavelength to blink. Press ▼, ▲ to adjust the blinking digit up or down and ◀, ▶ to change which digit blinks. Press the ENTER a second time to accept the new wavelength and exit the wavelength edit/display mode. Press λ or ESC to exit without changing the wavelength.

2.3.11 RANGE, Signal Range

RANGE key allows the user to view the amplifier signal range. Signal ranges step in 1 or 1/2 decade gain increments (MODE dependent) as the RANGE is changed in order to utilize maximum resolution from the meter's analog-to-digital converters. The available signal ranges are detector and mode dependent. Press RANGE to display the signal range number in the message display area. If AUTO is on, then the signal range number will change if the detector signal varies more than a decade in magnitude. Signal range changes can occur without an effect on the displayed measurement value. Press RANGE a second time to exit the signal range display mode.

Press the ▼, ▲ arrow keys to increase or decrease the signal range. If AUTO is on, pressing the ▼, ▲ arrow keys will disable AUTO and cause the signal range to change. Manual ranging is useful when working with external analog recording equipment.

NOTE

Pressing the ▼, ▲ arrow keys will disable AUTO ranging and change the signal range even when the signal range is not being displayed via the RANGE key.

2.3.12 R/S, Run-Stop

The R/S run-stop key provides front panel control over data acquisition for both channels simultaneously. If the channel is in SNGL measurement mode, each R/S key press causes one reading to be taken. In CONT measurement mode, each R/S key press toggles continuous data acquisition on and off.

The activity annunciator (Figure 2b) flashes to indicate that readings are being taken on the displayed channel as indicated in the upper right hand corner of the display. A steady glow indicates that the meter is armed and waiting for a pulse or P-P waveform to arrive. The indicator is off when data acquisition on the displayed channel has been stopped. In the special case that both channels are started together in DC CONT mode the readings will be synchronized together with channel A readings leading channel B readings by about 70 μ sec. Otherwise the channels run independent of each other.

2.3.13 MODE, Measurement Mode

The Model 2835-C provides a number of measurement modes for acquiring data. At power on, the meter checks the detector's calibration module to determine which measurement modes the detector supports. Table 3 describes the measurement modes available for each detector family.

Table 3 – Newport Detector Families and Available Measurement Modes

	Low-Power	High-Power	Energy
DC CONT	●	●	
DC SNGL	○	○	
INTG	○	○	
P-P CONT	○		
P-P SNGL	○		
CONT PULSE			●
SNGL PULSE			○

● marks the default mode for the detector family.

Press the MODE key repeatedly until the desired mode is indicated in the mode annunciator area of the display, see Table 4. When selecting a new mode, the display units will change to appropriate default units. See Section 2.3.14 for more details about units.

Table 4 – Model 2835-C Measurement Modes

Mode	Description
DC CONT	Measurement occurs at a programmable sample rate, Section 2.4.7.
DC SNGL	A measurement is taken every time the meter receives a trigger up to a 500Hz rate when two detector calibration modules are plugged into the meter and 1000Hz for one. (AUTO is disabled in SNGL mode.)
INTG	Measurements occur at 400Hz and are trapezoidally integrated to get an energy result. An R/S key press or external trigger sets the display to 0.0000 and sampling starts. A second R/S keypress or external trigger terminates integration.
P-P CONT	Acquisition is driven by the arrival of high-low peak pairs. A measurement is processed for every high-low peak pair up to a frequency of 500Hz when two detector calibration modules are plugged into the meter and 1000Hz for one. NOTE: pair captures experience a minimum 2 or 1 ms separation respectively.
P-P SNGL	A trigger ¹ arms the meter to capture the next high-low peak pair. Triggers can occur at a rate of up to 500Hz when two detector calibration modules are plugged into the meter and 1000Hz for one. (AUTO is disabled in SNGL mode.)
CONT PULSE	The meter captures every energy pulse up to a frequency of 500Hz when two detector calibration modules are plugged into the meter and 1000Hz for one. Above this rate, pulse acquisitions experience a minimum 2 or 1 ms separation respectively.
SNGL PULSE	A trigger ¹ arms the meter to capture the next energy pulse. Triggers can occur at a rate of up to 500Hz when two detector calibration modules are plugged into the meter and 1000Hz for one. (AUTO is disabled in SNGL mode.)

¹ “Trigger” refers to a command to start or stop signal acquisition. Trigger sources are the R/S key, the external trigger input and remote interface commands.

2.3.14 UNITS, Display Units

Measurements can be displayed in various units. The set of units available at any given time is determined by the detector type and the measurement mode. Press the UNITS key repeatedly to cycle the display through the set of available units. Table 5 describes the sets of available units for each detector family as a function of the measurement mode.

Table 5 – Valid Display Units Available to Detector Families by MODE.

FAMILY	MODE	V	A	W	W/cm ²	J	J/cm ²	Erg	Erg/cm ²	dBm	dB	REL
Low-Power	DC CONT		○	●	○					○	○	○
Low-Power	DC SNGL		○	●	○					○	○	○
Low-Power	INTG					●	○	○	○			
Low-Power	P-P CONT		○	●	○					○	○	○
Low-Power	P-P SNGL		○	●	○					○	○	○
High-Power	DC CONT	○		●	○					○	○	○
High-Power	DC SNGL	○		●	○					○	○	○
High-Power	INTG					●	○	○	○			
Energy	CONT PULSE	○				●	○	○	○			○
Energy	SNGL PULSE	○				●	○	○	○			○

● denotes the default units for the detector family in the given measurement mode. Some display units are abbreviated. Table 6 lists displayed units versus actual measurement units. Display units are limited to four characters in order to provide for display engineering prefixes such as: p, n, μ , m and k (pico, nano, micro, milli and kilo respectively).

Table 6 – Displayed Unit Abbreviations Versus Actual Measurement Units.

Actual Units	V	A	W	J	Erg	W/cm ²	J/cm ²	Erg/cm ²	dBm	dB	REL
Displayed Units	V	A	W	J	ERG	W/cm	J/cm	E/cm	dBm	dB	E±dd

2.3.15 STATS, Moving Statistics

The STATS key causes a list of statistical results from the stats buffer of the display channel to be displayed in the message display area. The stats buffer is a moving data window containing the most recent measurements to a depth $1 \leq N \leq 100$. The default value is $N = 10$. Display occurs without disrupting data acquisition or storage and results are continuously updated.

Press the STATS key to enter the stats display list. The first statistic displayed will be the stats buffer depth N. Press the ▼, ▲ adjust keys to move through the list. The following statistics are available: N, MAX, MIN, MAX-MIN, MEAN and STD DEV.

When the STATS buffer depth, N, is displayed, it can be adjusted by pressing ENTER and using the ▼, ▲ keys to set a new value for N. Press the ENTER key to adopt the new value for N and escape the edit mode. The STATS buffer is cleared whenever a new stats buffer depth N, or MODE or UNITS is established. Press the STATS key a second time to exit the stats display mode.

2.3.16 R/S A(B), Run-Stop Channel A (or B)

The R/S A and R/S B run-stop keys provides front panel control over data acquisition for each individual channel. When in SNGL measurement mode, each R/S A(B) key press causes one reading to be taken on the respective channel. In CONT measurement mode, each R/S A(B) key press toggles continuous data acquisition on and off on the respective channel.

The activity annunciator (Figure 2b) flashes to indicate that readings are being taken on the display channel as indicated in the upper righthand corner of the display. A steady glow indicates that the meter is armed and waiting for a pulse or P-P waveform to arrive. The indicator is off when data acquisition on the displayed channel has been stopped.

2.3.17 EXT, External Trigger

EXT enables and disables the triggering of data acquisition through the rear panel trigger input BNC connector. The meter can be configured to be triggered by either a rising or falling edge TTL signal. See Section 2.4.9.

Press EXT to light the EXT annunciator and enable the external trigger input to both channels. The Model 2835-C will still respond to triggers from the R/S, R/S A, or R/S B keys even when the EXT trigger is enabled. Press the EXT key a second time to turn off the annunciator and disable the external trigger input.

2.3.18 MENU

The MENU key provides access to advanced features and parameters. Press the MENU key to stop all data acquisition, light the MENU annunciator, and display the first item of the top level menu list. Press the MENU key a second time to immediately exit the menu. Menu structure and functions are presented in Section 2.4. The menu consists of a series of lists and parameter values. Most parameter values can be edited to configure the meter. Lists are moved through via the ▼, ▲ keys. To move to a lower level list, press the ENTER key. To escape a lower level list, press the ESC key.

In the menu, all key functions are ignored except for the MENU, ENTER, ESC and the ▼, ▲ keys. Each key will act without having to initially press SHIFT. Pressing SHIFT will toggle the SHIFT annunciator, but it will not effect anything else when in MENU mode.

To edit a displayed parameter press ENTER. This enables the edit mode and causes the parameter (or a digit) to blink. Press the ▼, ▲ keys to adjust the parameter through its allowed list or to count the blinking digit up and down. When adjusting a numerical value, the ◀, ▶ keys can be used to change which decimal digit blinks. Press ENTER to adopt the new value and escape the editing mode. Press ESC to leave the editing mode without adopting the new value.

2.3.19 ENTER

ENTER allows one to move lower into the menu when in menu mode, to enter editing mode when an editable parameter is being displayed and to adopt a new parameter value after it has been edited. Editable parameters can be found both in and out of the menu mode. If ENTER is pressed when a non-editable parameter is being displayed, the key press is ignored.

2.3.20 ESC, Escape

ESC allows one to escape to the next higher level when in menu mode and to escape editing mode when editing a parameter value without adopting any changes to the parameter. At the top level of the menu, pressing the ESC key exits the menu mode. The ESC key also allows one to escape various other display modes such as STATS or wavelength display or amplifier signal gain range display modes. When there is nothing to escape from, the ESC key is ignored.

2.3.21 ▼,▲,◀,▶ Adjust

The ▼,▲,◀,▶ keys allow one to adjust various parameter states and values. In normal operation, the ▼,▲ keys turn AUTO off and adjust the amplifier signal gain. In any other mode, the ▼,▲ keys cause one to move through a list or to adjust a blinking digit. The ◀,▶ keys zoom the bargraph (◀ in,▶ out) and in edit mode, select which digit of a numerical value will blink.

2.4 Menu Level Functions

Menu functions provide control over parameter values and methods of making measurements. The menu consists of a number of lists and parameter values. Table 7 presents a quick summary of the hierarchy of the menu functions and parameters. Most settings displayed in the menu and changes to those settings affect the channel indicated by the display channel annunciator.

2.4.1 Menu Access and Movement

To access the menu press MENU. The MENU annunciator will light and data acquisition will stop. Menu labels and parameter values are displayed in the message display area. The first, second and third leftmost decimal points within this area indicate a current position at the top, second, third or fourth level of the menu hierarchy, Figure 4. Table 7 presents the menu functions and their hierarchy.

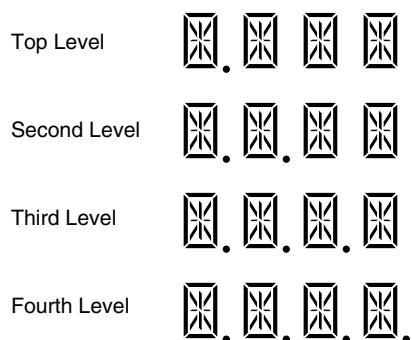


Figure 4 – Decimal Point Indication of Menu Hierarchy Position

In the menu, valid keys are limited to those needed to perform operations to move through the menu or edit a parameter value. In the menu, it is not necessary to hit the SHIFT key to invoke the action of the menu keys: MENU, ENTER, ESC or ▼,▲,◀,▶. The SHIFT key will toggle the SHIFT annunciator, but with no effect on subsequent pressings of other keys. Once in the menu, the rules for moving through the menu are as follows:

- Press ENTER to move to the next level down or to enter the edit mode if a editable parameter is being displayed and to accept a parameter value in its currently displayed state and exit edit mode.
- Press ESC to move to the next level up or to escape the menu when at the top level or to exit edit mode without changing the parameter value being edited.
- Press MENU to immediately escape the menu regardless of current level within the menu.

- iv. Use the ▼, ▲ keys to move up or down through a menu list on a given level. Also use these keys to adjust a parameter or decimal value when in edit mode.
- v. Use the ◀, ▶ keys to select the digit being adjusted when in edit mode.

Table 7 – Menu Level Key Functions and Parameters

Top Level		Second Level		Third Level
DATASTORE	<- ->	DATASTORE OFF CLR DATABUFFER (or DATABUFFER CLR) SLIDE BUFFER D_BUF SIZE dddd VIEW DATA	<- -> <- -> <- -> <- -> <- ->	DATASTORE ON, OFF SLIDE, FIX BUFFER D_BUF_SIZE dddd dddd OF dddd
SAVE CONFIG	<- ->	SAVE TO d		
RECALL CONFIG	<- ->	RCL DEFAULT, d		
ATTENUATOR	<- ->	ATTN ON, OFF		
AUTO CAL				
DET SWITCH POS	<- ->	SWITCH POS S, I, L		
USER CALIB	<- ->	USR CAL OFF USR RESP PRESENT RESP	<- -> <- -> <- ->	USR CAL ON, OFF d.dddE±d A/W [†] d.dddE±d A/W [†]
DC SAMPLING	<- ->	SAMPLE PREC SAMPLE FREQ	<- -> <- ->	PREC = 20000, 4096 CNT FREQ = ddd.ddd HZ
TRIGGER OUT	<- ->	TRIG ON CMPLT TRIG POLARITY TRIGGER FREQ TRIGGER LEVEL TRIGGER INTG	<- -> <- -> <- -> <- ->	TRIG ON CMPLT , TRIG AT FREQ , TRIG ON LEVEL , TRIG ON INTG , TRIG OUT OFF TRIG ACTIVE LO, HI FREQ = dddd.ddd HZ LVL = ±d.ddd E±dd W INTG = ±d.ddd E±dd J
EXT TRIGGER IN	<- ->	FALLING, RISING EDGE		
BAR GRAPH ON	<- ->	BAR GRAPH ON, OFF		
tone OFF	<- ->	tone ON, OFF		
REMOTE SETUP	<- ->	GPIB ADDR dd BAUD RATE 9600 RS-232 ECHO OFF	<- -> <- -> <- ->	GPIB ADDR dd BAUD RATE 1200, 2400, 4800, 9600, 19.2K RS-232 ECHO ON, OFF
GENERAL INFO	<- ->	MODEL 2835-C SW VERSION d.d DETECTOR INFO	<- ->	MODEL xxxxxx DET SN ddddd ATTN SN ddddd CAL ddMONyyyy

[†] Units may be in A/W, V/W or V/J.

Items in ***bold italics*** type are editable states or decimal values. Decimals are denoted by ***d***. Move vertically via the ▼, ▲ keys. Move horizontally via ENTER and ESC.

2.4.2 Data Store

The Model 2835-C allows a user to save up to 1000 measurements for each channel for a total of 2000 measurements. These measurements are stored in an internal buffer for subsequent viewing or transmission over a computer interface. A separate buffer is maintained and configured for each channel. Data is maintained on power down, but lost when a new configuration is loaded, Section 2.4.3, or when the buffer is cleared via the CLR DATA BUFFER command or when data with new units is being stored.

The data store buffer operates in two ways: SLIDE or FIX. In SLIDE configuration, the buffer slides along storing the most recent measurements up to the size of the buffer. Beyond this, as data enters the buffer, the oldest data is pushed out and lost. In FIX configuration, data storing continues until the data buffer is full. After this, data acquisition stops and no additional data can be stored without first clearing the buffer via the CLR DATA BUFFER command. DATA BUFFER CLR is displayed when the buffer is empty.

The size of the buffer is set by the D_BUF_SIZE ***dddd*** menu function. Edit the value ***dddd*** to establish the number of data points that the buffer will hold before dropping old data or stopping data storage.

Data storing is enabled by the user via the DATA STORE menu function. Edit the ***ON, OFF*** condition to enable or disable data storing and the associated STORE annunciator.

When in CONT mode, data acquisition and storage is resumed immediately upon exiting the menus if acquisition is active as the menu is entered. When acquisition is not active when the menu is entered, or when in SNGL or INTG mode, data acquisition and storage requires an initiating R/S key press, external trigger or a remote RUN command upon exiting the menu.

Buffer data can be viewed via the VIEW DATA menu command. Data are displayed in the measurement area while the message area displays the position within the buffer: ***dddd*** of ***dddd***. Use the ▼, ▲ keys to move through the buffer data. The value ***dddd*** = 0001 is the first, i.e. the oldest datum.

NOTE

When DATA STORE is off, CONT acquisition mode behavior defaults to the condition where data acquisition is begun without the requirement of a starting trigger. SNGL acquisition modes always require a trigger for each acquisition.

Table 8 – Data Store Operations

Menu Operation	Keypad Commands	Associated Remote Commands
DATA STORE <i>ON, OFF</i>	Edit <i>ON, OFF</i> status.	DSE_n, DSE_n?
DATA BUFFER CLR or CLR DATA BUFFER	If CLR DATA BUFFER, press ENTER to clear buffer	DSCLR_n
<i>SLIDE, FIX</i> BUFFER	Edit <i>SLIDE, FIX</i> BUFFER	DSBUF_n, DSBUF_n?
D_BUF_SIZE <i>dddd</i>	Edit D_BUF_SIZE <i>dddd</i>	DSSZ_n, DSSZ_n?
VIEW DATA	Press ENTER and use ▼, ▲ keys.	DS_n?, DSCNT_n?, DSUNITS_n?

NOTE

n in a remote command stands for A (channel A) or B (channel B).

2.4.3 Meter Configuration

The Model 2835-C provides a method to save the configuration of the entire meter and to recall that configuration for later use even if the meter has been turned off. This is accomplished through configuration buffers maintained in nonvolatile memory. Configuration buffers are numbered 0 to 9 with buffer 0 being a DEFAULT buffer which can only be recalled but not saved to. The reset state of all the buffers except the default buffer is empty. Empty buffers cannot be recalled.

A recalled configuration becomes the current configuration of the meter. Any changes to the current configuration must be saved via SAVE CONFIG or they will be lost when a new configuration is recalled.

The meter will not recall a configuration that is not compatible with either detector calibration module currently plugged into the meter. Configurations using the same model of detector are compatible while configurations using different models of detectors are incompatible. The default configuration of the meter depends upon the detector family. The list of configuration parameters stored in a configuration buffer as well as their default values versus detector family are listed in Table 9 below:

Table 9 – Configuration Parameters and Default Conditions

Parameter	Detector Family	Default Condition
MODE	Low-Power, High-Power Energy	DC CONT CONT PULSE
UNITS	Low-Power, High-Power Energy	W J
λ , Lambda		Lowest available
PRESENT RESP		PRESENT RESP
USRCAL		OFF
ATTN		OFF
FILTER		OFF
DC SAMPLE PREC	Low-Power, High-Power	20,000 CNT
DC SAMPLE FREQ		25 Hz
AUTO		ON
RANGE		Lowest available
ZERO		OFF
Zero Value		0.000
REF SEL		STO REF
Reference Value		0.001
STATS Buffer Size		10
DET SWITCH POS	Energy	I, (Intermediate)
DATA STORE		OFF
DATA STORE BUFFER	SLIDE	

Parameter	Detector Family	Default Condition
D_BUF_SIZE		100
Data Store Units		Same as UNITS
EXT		OFF
EXT TRIG IN		FALLING
TRIGGER OUT		TRIG ON CMPLT
TRIG OUT POL		TRIG ACTIVE LO
TRIG OUT FREQ		FREQ = 30 Hz
TRIG LEVEL	Low-Power, High-Power Energy	LVL = 0.001 W LVL = 0.001 J
TRIG INTG LVL	Low-Power, High-Power	INTG = 0.001 J
BAR GRAPH		ON
TONE		OFF
GPIB ADDR		05
BAUD RATE		9600
RS-232 ECHO		OFF

These parameters adopt the following default values at power up and are not affected by recalling a configuration or by setting the configuration to default.

Local Lockout	OFF
Display Brightness	NORMAL

Table 10. below lists the menu commands effecting the saving and recalling of meter configurations.

Table 10 – Meter Configuration Operations

Menu Operation	Keypad Commands	Associated Remote Commands
SAVE CONFIG	Adjust SAVE TO d , and ENTER	*SAV
RECALL CONFIG	Adjust RECALL d , and ENTER	*RCL, *RST

2.4.4 AUTO CAL

The AUTO CAL command causes the 2835-C to perform A/D conversions of amplifier offset voltages (zero errors) arising from aging and temperature effects. These conversions are then used in subtracting the appropriate error voltage from each reading during normal operation. The 2835-C automatically performs this procedure every time it powers up (or is reset). To achieve stable reading at the specified accuracy, AUTO CAL should be executed for each channel after a minimum 60 minute warm-up period from power-up.

Executing AUTO CAL with High Power (Thermopile) detectors:

1. With the detector connected to the 2835-C, remove the detector from the radiation source and allow a minimum of 60 seconds for the detector surface temperature to stabilize.

2. Press ENTER when AUTO CAL is displayed. The display message area will display “ONE MOMENT”, followed by a buzzer sound indicating that AUTO CAL is complete.

NOTE:

Although at Power up, an AUTOCAL is performed, the above method is necessary for proper calibration when the 2835-C is used with voltage sources.

Executing AUTO CAL with Low Power (Photodiode) or Energy detectors:

Simply press ENTER when AUTO CAL is displayed. The display message area will display “ONE MOMENT”, followed by a buzzer sound indicating that AUTO CAL is complete. Low Power and Energy detectors do not have to be connected to the 2835-C or removed from the radiation source to effectively execute AUTO CAL.

2.4.5 Attenuator

The Attenuator selects the responsivity value, R_λ , to be the value for the detector alone or the value for the detector-with-attenuator. When ATTN is ON the ATTN annunciator is lit and the responsivity of the detector-with-attenuator is used. When ATTN is OFF, the annunciator is off and the detector-alone responsivity is used. If the detector does not have an attenuator, or if USR CAL is on (Section 2.4.6) the ATTN setting has no effect.

2.4.6 User Calibration

The Model 2835-C allows one to create a detector responsivity which overrides the responsivities obtained from the detector’s calibration module. This allows one to account for the effects of additional optics and filters in the measurement path. When USR CAL is on, the USR CAL annunciator is lit and the meter adopts the responsivity value displayed by the editable USR RESP value. Use the PRESENT RESP function to display the current calibration module responsivity.

The USR RESP units are the same as the PRESENT RESP units. Table 11 lists the possible user calibration operations.

Table 11 – User Calibration Operations

Menu Operation	Keypad Commands	Associated Remote Commands
USR CALIB ON, OFF	Edit USR CALIB ON, OFF	USRCAL_n, USRCAL_n?
USR RESP	Edit d.dddE±dd A/W*	USRRESP_n, USERRESP_n?
PRESENT RESP	ENTER to view d.dddE±dd A/W*	RESP_n?

*Units of A/W, V/W or V/J may be displayed. These units are not editable.

NOTE

n in a remote command stands for A (channel A) or B (channel B).

2.4.7 DC Sampling

The Model 2835-C incorporates two analog-to-digital, (A/D) converters, one with 20,000 count resolution and a second with 4096 count resolution. The user may select which A/D will be used for both channels during DC CONT and DC SNGL acquisition modes. All other modes use the 4096 count A/D. The 20,000 count A/D converter can operate at sample rates up to 25 Hz. When two detector calibration modules are plugged into the meter, the 4,096 count A/D converter can operate at sample rates up to 500 Hz. When one detector calibration module is plugged into the meter, the 4,096 count A/D converter can operate at sample rates up to 1000 Hz.

The SAMPLE PREC menu command, Table 12, selects which analog-to-digital converter is used for both channels. The sampling frequency for both channels can be adjusted within the limits imposed by the SAMPLE PREC state, see Table 13. When the SAMPLE PREC state changes, the sampling frequency defaults to 25 Hz if the existing SAMPLE FREQ is incompatible with the new SAMPLE PREC state. The SAMPLE FREQ and the SAMPLE PREC settings affect both channels, not just the the display channel.

Table 12 – SAMPLE PREC States and Limits.

SAMPLE PREC	A/D Accuracy	Sample Frequency Range
20000 CNT	20,000 counts	0.001 Hz to 25.0 Hz
4096 CNT	4,096 counts	0.001 Hz to 500.0 Hz or 1000.0 Hz

Table 13 – DC SAMPLING Operations.

Menu Function	Keypad Commands	Associated Remote Commands
SAMPLE PREC	Edit PREC= 20,000 , 4096 CNT	SPREC,SPREC?
SAMPLE FREQ	Edit FREQ= ddd.ddd HZ	SFREQ,SFREQ?

2.4.8 Trigger Output

The Model 2835-C's rear panel TTL trigger output allows it to coordinate activities among other instruments. The trigger output can operate in several ways: conversion complete, periodic output, comparator output and integrating comparator output. In addition, the polarity of the trigger output can be specified.

In TRIG ON CMPLT mode, a pulse is output after each reading has been acquired and completely processed from either channel. When acquiring on one channel, it indicates that the 2835-C is ready to take another reading. The width of this pulse is at least 8 μ S.

In TRIG AT FREQ mode, a pulse is output at a user defined frequency or rate. Each pulse width is at least 8 μ S and the programmable frequency range of the pulses is 0.001 Hz to 1000.0 Hz.

In TRIG ON LEVEL mode, each measurement is compared to a programmable trigger level. The units of the trigger level always equal Watts for power detectors or Joules for energy detectors. If a measurement is less than the specified value, then the trigger output is inactive. If the measurement exceeds the specified value then the trigger output becomes active. The TRIG ON LEVEL output is a shift in level rather than a pulse.

In TRIG ON INTG mode, continuous measurements from a power detector are integrated and compared against a programmable value. The units of the programmable value are Joules. If the calculated integral is less than the specified value, then the trigger output is active. If the calculated integral exceeds the specified value, then the trigger output becomes inactive. The TRIG ON INTG trigger is a shift in level rather than a pulse.

If both channels are sampling when this function is initiated, the time lag between subsequent triggers is less than 700 μ sec. If a R/S A or R/S B is initiated twice, then the separation between TTL OUTPUTS is relative to the RUN command initialization, giving a time lag up to one sample period between channel triggers (depending on sample frequency chosen).

The polarity of the trigger output is programmable as active high or active low. If the polarity is active high then the output will idle low. If the polarity is active low then the line will idle high.

Table 14 – TRIGGER OUT Operations.

Menu Function	Keypad Commands	Associated Remote Commands
TRIGGEROUT	Edit TRIG ON CMPLT TRIG AT FREQ TRIG ON LEVEL TRIG ON INTG TRIG OUT OFF	TRIGOUT, TRIGOUT?
TRIGPOLARITY	Edit ACTIVE HI, LO	TRIGOUTPOL, TRIGOUTPOL?
TRIGGERFREQ	Edit FREQ= ddd.ddd Hz	TRIGOUTFREQ, TRIGOUTFREQ?
TRIGGERLEVEL	Edit LVL= d.dddE$\pm$dd W*	TRIGOUTLVL, TRIGOUTLVL?
TRIGGERINTG	Edit INTG= d.dddE$\pm$dd J	TRIGOUTINTG, TRIGOUTINTG?

*Displayed units may be W or J depending upon the detector in use.

2.4.9 Trigger Input

The Model 2835-C's rear panel TTL external trigger input can be enabled or disabled, Section 2.3.17, and have its edge polarity set. The edge polarity is accessed via the EXT TRIGGER IN menu function and can be set to rising or falling edge triggering.

The external trigger affects both channels and, like the R/S key, acts like an acquisition trigger when in SNGL modes and acts like a toggled acquisition gate when in CONT modes. In Peak-to-Peak or Pulse acquisition modes the external trigger (and R/S key) act to enable single or continuous acquisition rather than forcing an acquisition to occur at the moment the trigger (or keypress) occurs.

The external trigger pulse width must be ≥ 200 ns. In DC SNGL mode, the delay from the trigger input going active to the start of the A/D conversion is $< 80 \mu$ s. In SNGL PULSE and P-P SNGL modes, the delay from the trigger input going active to the meter becoming armed to capture the next PULSE or P-P pair is $< 70 \mu$ s.

Table 15 – EXT TRIGGER IN Operations.

Menu Function	Keypad Commands	Associated Remote Commands
EXT TRIGGER IN	Edit RISING, FALLING EDGE	EXTEDGE, EXTEDGE?, EXT, EXT?

2.4.10 Bar Graph

A high resolution analog bar graph is provided for visual feedback when performing optical alignment tasks. This bar graph uses the message display area and moves from left to right with increasing display value, and reaches full scale when the digital display reaches full scale.

The ◀, ▶ keys can be used to zoom the bargraph in and out. Press ◀ to zoom the bargraph into a 10x magnification which auto-centers with each subsequent ◀ press. Press ▶ to zoom out to normal 1x magnification.

Table 16. BARGRAPH Operations.

Menu Function	Keypad Commands	Associated Remote Commands
BARGRAPH	Edit BARGRAPH ON, OFF	BARGRAPH, BARGRAPH?

2.4.11 Tone

The 2835-C has a speaker that can emit a short tone or tick to indicate when a peak or pulse is detected in P-P or PULSE measurement modes for the display channel. The audible output is enabled or disabled via the TONE menu function. TONE is not available in INTG measurement mode.

Table 17 – TONE Operations.

Menu Function	Keypad Commands	Associated Remote Commands
TONE	Edit TONE ON, OFF	TONE, TONE?

2.4.12 Detector Switch Position

Some Newport Energy detectors have a switch that sets an electronic decay time constant to optimize operation for various pulse repetition rates. As this adjustment effects the responsivity of the detector, the Model 2835-C must be informed of the detector's switch setting. The available detector switch settings are: SHORT, INTERMEDIATE and LONG. The 2835-C accounts for the switch setting through the DET SWITCH POS function.

Table 18 – DET SWITCH POS Operations.

Menu Function	Keypad Commands	Associated Remote Commands
DET SWITCH POS	Edit SWITCH POS S, I, L	DETSW DETSW?

2.4.13 Remote Setup

The Model 2835-C provides both RS-232 and IEEE-488 computer interfaces as standard features. Each interface requires that certain parameters be set. Remote interface setup and commands are discussed in Section 4, Computer Interfacing.

Table 19 – REMOTE SETUP Operations.

Menu Function	Keypad Commands	Associated Remote Commands
GPB ADDR dd	Edit GPB ADDR dd	None
BAUD RATE dddd	Edit BAUD RATE 1200, 2400, 4800, 9600, 19.2K	None
RS-232 ECHO ON, OFF	Edit RS-232 ECHO ON, OFF	ECHO, ECHO?

2.4.14 General Information Functions

General information about the Model 2835-C and any attached detector can be found using the GENERAL INFO menu functions. These functions display the Model 2835-C software version, the detector's model number, serial number, associated attenuator serial number and calibration date.

Table 20 – GENERAL INFO Operations.

Menu Function	Keypad Commands	Associated Remote Commands
MODEL ccccccc	For viewing only.	DETMODEL?
SW VERSION d.d	For viewing only.	*IDN?
DET SN dddddd	For viewing only.	DETSN?
ATTN SN dddddd	For viewing only.	ATTNSN?
CAL ddMonyyyy	For viewing only.	CALDATE?

2.5 Connecting AC Power

Model 2835-C can be configured to operate with line voltages over the following ranges: 90-264 VAC, 50-60 Hz. Before turning the meter on, configure it to local voltage using the following procedure:

- Configure the Model 2835-C's power supply voltage selection switches to match the nominal local voltage. See Figure 5 and Table 21.
- Plug an AC line power cord to the rear of the Model 2835-C and then connect the cord with AC power.

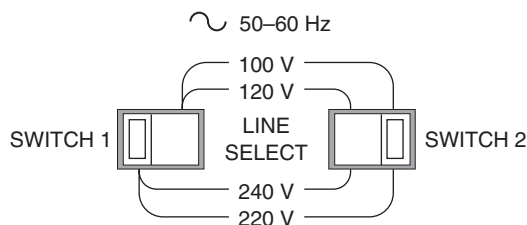


Figure 5 – Rear Panel Power Supply Voltage Switches in Positions with Switch 1 set to Left and Switch 2 set to Right

Table 21 – Power Supply Voltage Switch Positions.

Switch 1 Position	Switch 2 Position	Nominal Local Voltage
Right	Right	100 VAC
Right	Left	120 VAC
Left	Right	220 VAC
Left	Left	240 VAC

WARNING

This product is equipped with a 3 wire grounding type plug. Any interruption of the grounding connection can create an electric shock hazard.

If you are unable to insert the plug into your wall plug receptacle, contact your electrician to perform the necessary alterations to assure that the green (green-yellow) wire is attached to earth ground.

2.6 Detector Connection and Setup

Connect the detector to its calibration module as shown, Figure 6. The detector's model and serial numbers should match the model and serial numbers found on the calibration module. Insert the calibration module into the front panel input port of the Model 2835-C. An alignment pin forces the proper orientation of the calibration module.

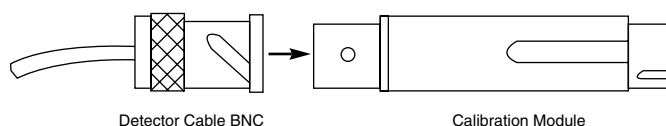


Figure 6 – Connecting a Detector with its Calibration Module

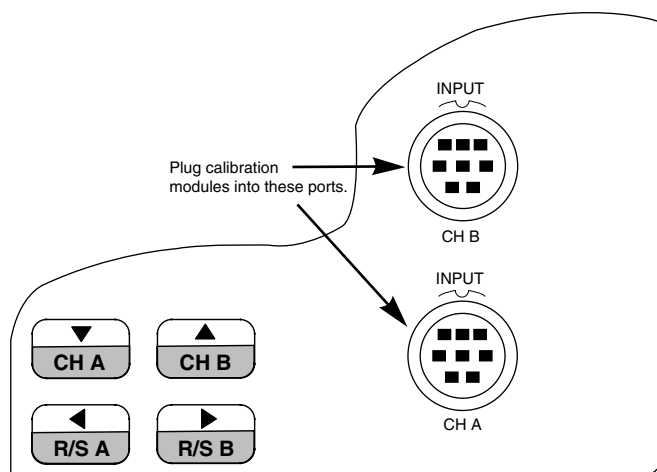


Figure 7 – Model 2835-C Detector Calibration Module Input Port

NOTE

If the detector being connected is a battery powered Newport Energy Detector then be sure to configure the DET SWITCH POS menu function to match the position of the switch on the detector, see Section 2.4.12. The 2835-C defaults to the intermediate switch position, I.

2.7 PowerUp

Turn on the Model 2835-C by depressing the front panel key until it clicks in and remains in its depressed position. The Model 2835-C will perform a power up self test and then configure itself to its last operating state. If the last operating state is not compatible with one of the current detectors, the 2835-C adopts the default configuration appropriate to this new detector. If the meter does not pass its self test or fails to respond to front panel key commands, refer to Section 6, Maintenance, Test and Troubleshooting.

2.8 Performing Basic Measurements

Basic measurement techniques for using the Model 2835-C are covered in the following sections. Refer to Table 2, Table 3 and Table 7 (pages 6, 10, and 15 respectively) for a review of the Model 2835-C's functions and capabilities. The following instructions assume familiarity with the meters functions. They also include steps to incorporate background correction and assume that the experimental setup underfills and does not saturate or damage the detector.

2.8.1 Making DC Power Measurements

The following process describes the procedure for making basic optical power measurements while properly removing the influence of ambient light and other drift effects.

- i. Plug in a Newport Low-Power or High-Power detector via its associated calibration module and then turn the meter on. Set MODE to DC CONT, set AUTO on and set the measurement wavelength to the desired value.
- ii. When using a Newport High-Power (thermopile) detector, execute AUTO CAL per Section 2.4.4.
- iii. Cover or otherwise block the source that you will be measuring and then turn ZERO on.
- iv. Uncover the source so it illuminates the detector and note the display value. This reading is the optical power observed by the detector due to the source.

NOTE

The process as detailed assumes that the ambient signal is not changing between when you zero the display and when you make your measurement. Remember, if you can see your detector as you move around, then your detector can see you as a changing ambient DC signal!

2.8.2 Making Peak-to-Peak Power Measurements

The following process describes the procedure for making basic optical peak-to-peak power measurements.

- i. Plug in a Newport Low-Power detector via its associated calibration module and then turn the meter on. Set MODE to P-P CONT, set AUTO on and set the measurement wavelength to the desired value.
- ii. Illuminate the detector and note the display value. This reading is the peak-to-peak optical power observed by the detector.

2.8.3 Making Pulse Energy Measurements

The following process describes the procedure for making basic optical pulse energy measurements.

- i. Plug in a Newport Energy detector via its associated calibration module and then turn the meter on. Set MODE to CONT PULSE, set AUTO on and also set the measurement wavelength to the desired value.
- ii. Illuminate the detector and note the display value as the meter measures each laser pulse. These readings represent the energies of the incident laser pulses. The meter will display the last pulse energy measured until a new pulse arrives.

2.8.4 Making a Signal Integration Measurement

The following process describes the procedure for making a basic signal integration measurement while properly removing influence of ambient light and other drift effects. The Model 2835-C begins and ends integration via the arrival of a trigger signal such as is received from the R/S key.

- i. Plug in a Newport Low-Power or High-Power detector via its associated calibration module and then turn the meter on. Set MODE to DC CONT, set AUTO on and also set the measurement wavelength to the desired value.
- ii. When using a Newport High-Power (thermopile) detector, execute AUTO CAL per Section 2.4.4.
- iii. Cover or otherwise block the source that you will be measuring, turn ZERO on and then set the MODE to INTG. As you enter the INTG mode, the meter will begin to acquire and integrate data. The display value may reflect the integration of noise due to ambient temperature fluctuations (thermopile) or light fluctuations (photodiode).
- iv. Uncover or trigger the source. The display value should now reflect detector signal integration process by continuously increasing.
- v. Press the R/S key to stop the integration and freeze the display value at the final integration value.

NOTE

1. The process as described above assumes that ambient signals are not changing between when you zero the display and when you make your measurement.
2. The 2835-C will not integrate signals less than 10 pA. If your noise level is greater, the ZERO command may help (step iii).

2.8.5 Measuring a Laser Pulse Energy with a Thermopile Detector

This application makes use of the Model 2835-C's INTG mode. When an optical pulse with energy $E(\lambda)$ is incident on a thermopile, a voltage signal arises as the heat pulse flows out to the cooling fins. The integrated signal resulting from this heat pulse is a measure of the optical pulse energy. See Figure 8.

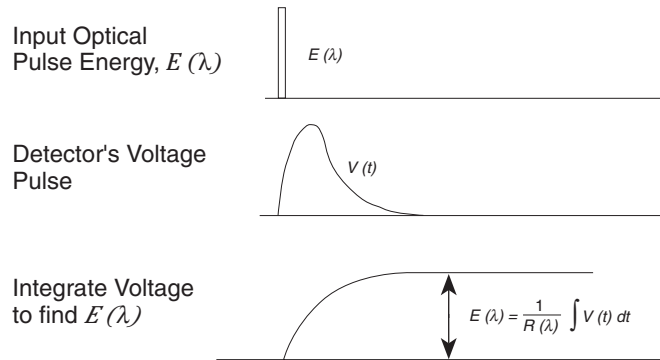


Figure 8 – Measuring Laser Pulse Energy via a Thermopile in INTG Mode

A recommended procedure is:

- i. Plug in a Newport High-Power detector via its associated calibration module and then turn the meter on. Set MODE to DC CONT, set AUTO on, set the measurement wavelength to the desired value and then execute AUTO CAL per Section 2.4.4. Now set ZERO on and then MODE to INTG.
- ii. Press the R/S key to begin the integration. Before the optical pulse arrives, the display may reflect the integral of detector noise due to ambient temperature fluctuations.
- iii. Trigger the laser pulse and watch the display increase at a much faster rate due to the detector signal from the heat flow.
- iv. When the display increase begins to slow down and again reflect integration of the detectors noise component, press the R/S key a second time to stop the integration. The final display value represents the pulse energy.

NOTE

If the integrated result of the pulse signal is much larger than the integral of the detector's noise component, the display will show a steady reading after the pulse. If the integrated result of the pulse is not much larger, then error in the measurement will arise due to the uncertainty generated by integration of the noise component terms.

NOTE

The time constant of a thermopile detector determines the amount of time that one should expect to wait when making an integrated energy measurement of an optical pulse. Typically, an accurate value will be arrived at 5 time constants after the arrival of the optical pulse. Newport High-Power detector time constants are listed in the Newport Catalog and in each detector's manual.

2.8.6 Using the Model 2835-C as an Exposure Controller

This application uses the INTG measurement mode along with triggering features to configure the Model 2835-C as an exposure meter-controller. Use the following setup and procedure:

- i. Connect a BNC cable between the Model 2835-C's trigger input and trigger output BNC connectors. Using a T-connector, connect the trigger output BNC to your shutter's external control BNC.

-
- ii. Configure the trigger output to TRIG AT INTG and set the energy level at which you wish the exposure to stop, see Section 2.4.7. Adjust the polarity of the trigger output to open the shutter when the trigger output is active.
 - iii. Set the meter to INTG mode with AUTO on and then press the R/S key to begin the integration and open the exposure controlling shutter.
 - iv. The display value will increase until the INTG trigger output threshold is crossed. At the threshold, integration will stop and the trigger output will close the shutter. The display now shows the final integration value from the exposure.

NOTE

You can also configure the Model 2835-C as an exposure timer-controller by configuring the trigger output to TRIG AT FREQ and setting the trigger frequency appropriately.

Section 3

Principles of Operation

3.1 Introduction

The Model 2835-C's electronics adapt to a number of signal measurement tasks: DC current or voltage, AC peak-to-peak current or pulse voltage, or integrated DC current or voltage signals. This versatility is required to handle the various signals that Newport's **Low-Power**, **High-Power** and **Energy** detector families generate. These detector families are based on semiconductor, thermopile and pyroelectric detectors respectively. Detector data is introduced to the Model 2835-C by way of a calibration module specific to the detector in use. At power up (and RESET), the 2835-C uploads information about the detector from the calibration module which describes the set of operating states available to the detector. A user then selects among the available operating states when using the meter. Front panel control and the operating states of the Model 2835-C are discussed in Sections 2.3 and 2.4.

3.2 Analog Signal Flow

Detector signals can follow many different paths through the Model 2835-C input amplifier chain. A block diagram of analog signal flow is shown in Figure 9. The actual flow path depends upon the detector type and the mode of measurement.

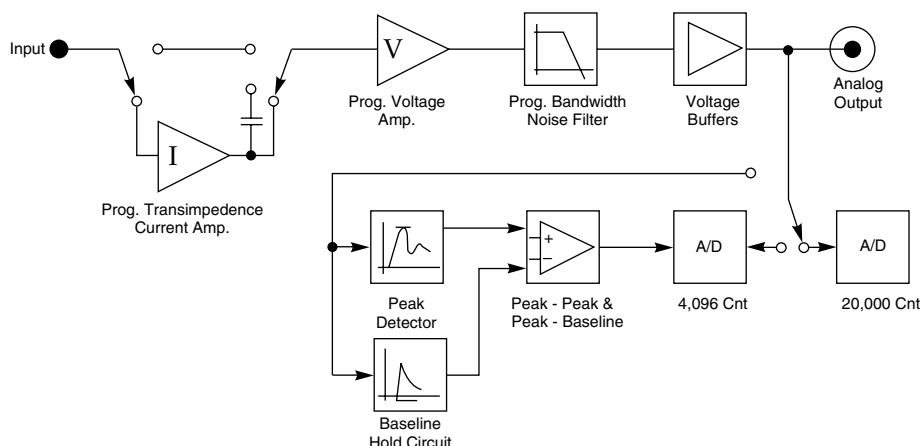


Figure 9 – Model 2835-C Analog Signal Flow Diagram

The analog signal flow path is primarily determined by the responsivity units of the detector. The numerator of these units indicates how the meter must be configured in order to obtain a calibrated optical measurement. Analog signal flow is independent of whether SNGL or CONT measurements are made. Responsivity units and signal flows for the various detector families are listed in Table 22.

Table 22 – Analog Signal Flow Paths.

Detector Family	Resp. Units	Mode	Amplifier Path	Peak-Peak	A/D Converter
Low-Power	A / W	DC	I	No	20,000 or 4,096
Low-Power	A / W	P-P	I, AC coupled	Yes	4,096
Low-Power	A / W	INTG	I	No	4,096
High-Power	V / W	DC	V only	No	20,000 or 4,096
High-Power	V / W	INTG	V only	No	4,096
Energy	V / J	PULSE	V only	Yes	4,096

3.3 Digitized Signal Flow

Signals that are captured by one of the analog-to-digital converters are further processed as illustrated in the signal flow block diagram of Figure 10. Raw signals are acquired by the analog-to-digital converters and have the units of current or voltage depending upon how the input amplifier chain was configured. These digitized signals move through a number of process steps which may or may not alter the digitized value depending upon the operating state of the meter. Each of these possible processing steps is discussed in their order of occurrence.

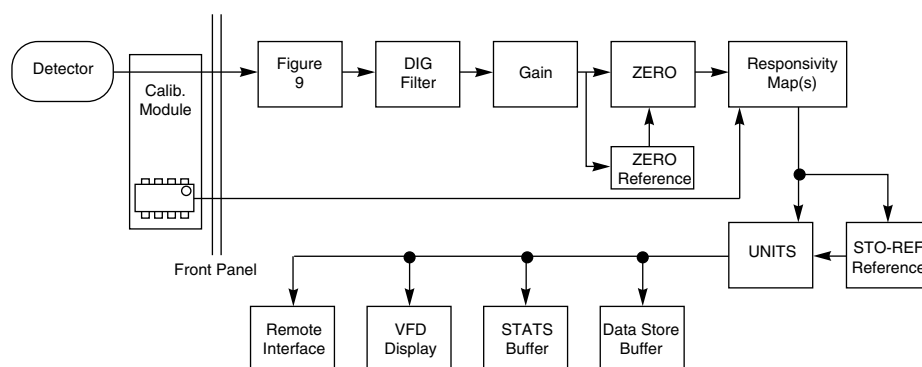


Figure 10 – Model 2835-C Digitized Signal Flow Block Diagram

Digital Filter

If the digital filter annunciator DIG is on, Section 2.3.4, the filter output is the average of the most recent 10 digitized values. When less than 10 values have been acquired since the last reset of the digital filter, the output is the average of all the values received. The digital filter is reset when the Model 2835-C is turned on and whenever the UNITS or the range changes or when the MODE key is pressed.

NOTE

When using the digital filter in SNGL acquisition mode, each measurement is the average of the last 10 acquisitions independent of how old any of the measurements are.

Gain

Gain processing accounts for the signal gain of the input amplifiers. The output is the product of the digitized value and the amplifier gain.

Zero Offset

Zero offset is active whenever the ZERO annunciator is lit, Section 2.3.5. The zero offset output is equal to the input value less the zero reference value, $S-S_0$.

Responsivity Map(s)

This process scales the input value in accordance with current calibration wavelength and the responsivity map downloaded from the detector calibration module. The output of this process, i.e. the measurement value, is the digitized input value divided by the responsivity associated with the current calibration wavelength or the user defined calibration value. Different responsivities are used depending upon if the attenuator, ATTN, annunciator is lit, Section 2.4.5.

Units Correction

Units correction adjusts a measurement value to account for the display units selected. When the display units are equal to the detector signal units, i.e. equal to the numerator of the responsivity units, Table 22, the measurement value is not adjusted. Otherwise the digitized value is adjusted to account for detector responsivity and/or additional unit conversions such as Joules-to-Ergs or W-to-W/cm².

NOTE

Per area unit conversions such as W-to-W/cm² divide the measurement value by the active area of the detector. This calculation assumes that the entire active area of the detector is uniformly illuminated. Per area measurements where the entire detector active area is not uniformly lit are not accurate. The user must insure that these conditions are met before utilizing per area units.

3.4 Typical Detector Signals

The flexibility of the Model 2835-C analog signal flow is required in order that it may properly measure the signals that various types of detectors make. Basic optical power or energy measurements are related to a measured detector signal, S , in the following way:

$$P \text{ or } E = \frac{S}{R_\lambda}$$

Where: R_λ = Detector responsivity at λ .
 S = Detector signal

Semiconductor (Newport Low-Power) detectors provide a current signal. The 2835-C is capable of 100fA resolution in order to provide the highest sensitivity performance with these detectors.

Thermopile (Newport High-Power) detectors, provide a small voltage signal. The 2835-C is capable of 125 nV resolution in order to reach the sensitivity limits of thermopile detectors.

Pyroelectric (Newport Energy) detectors, deliver a peak voltage signal, . The 2835-C is capable of capturing 2 μ sec rise time voltage spikes so that it may operated with the fastest of these detectors.

3.5 Thermopile Detector Signals

Thermopile detectors respond with a voltage signal that slowly to changes in incident optical power. The time constant of most thermopile detectors is on the order of 1 to 10 seconds.

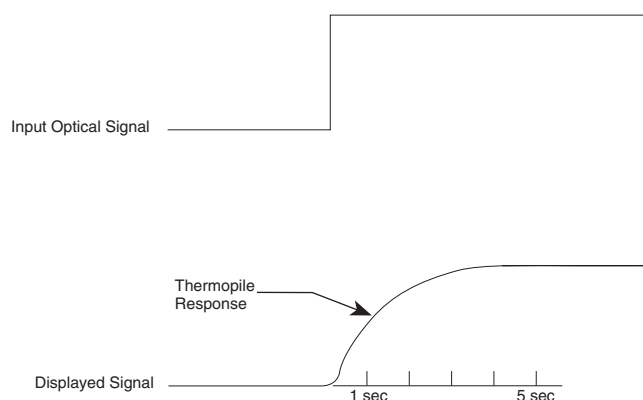


Figure 11 – Thermopile Signals exhibit 1 to 10 second time constants.

3.6 Pulse Energy Detector Signals

A Newport Energy detector will respond to a single radiant energy pulse with a voltage pulse at its BNC output. This pulse exhibits a sharp voltage rise to a peak followed by a slower voltage decay which “undershoots” zero volts before settling back to zero volts. When a detector is operated within its proper limits, the voltage difference from immediately before the sharp rise to the peak is linearly proportional to the radiant energy.

If a second pulse arrives before the “undershoot” rises back to zero volts, the voltage rise from this pulse will start from an initial negative value. At sufficient energy pulse repetition rates, a negative “baseline” voltage will develop from which the voltage rise must now be measured to achieve accurate energy readings. The 2835-C contains baseline capture circuitry which maintain its accuracy specifications over rep-rates ranging from single pulse to 2 KHz.

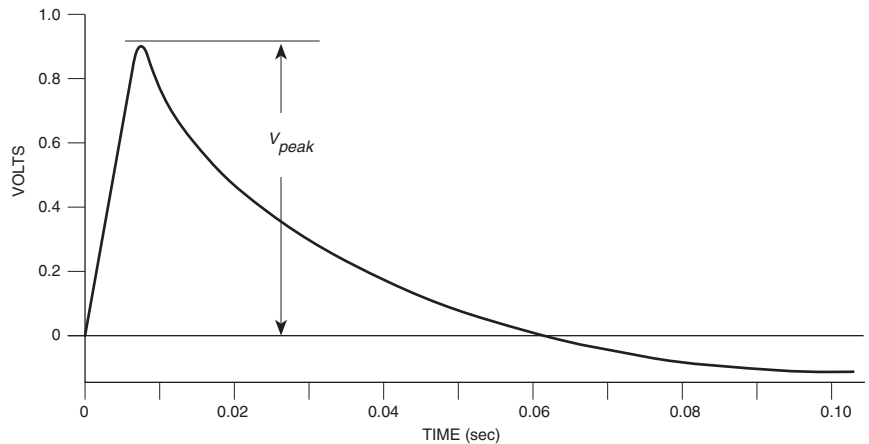


Figure 12 – Typical Newport Energy Detector Signal Waveform - An energy detector signal sharply rises to a peak value and then decays going somewhat negative before finally returning to zero. The energy in the radiant pulse is proportional to the height of the peak measured from immediately before the sharp rise.

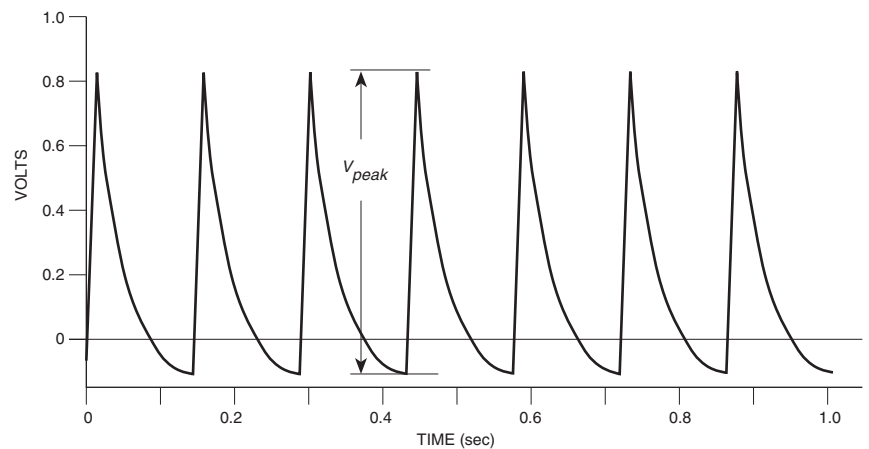


Figure 13 – Negative Baseline Voltage Due to a Pulse Train - If a laser pulse arrives before the previous Energy detector signal has fully decayed, the detector signal rises from the present decay point of the previous signal.

3.7 Peak-to-Peak (Photodiode) Detector Signals

The Model 2835-C allows one to make peak-to-peak measurements of time varying signals from semiconductor photodiode detectors. Since optical power is a zero bounded positive quantity, signals from a detector observing such modulated light will similarly be zero bounded positive signals. To make a peak-to-peak measurement, the Model 2835-C must be able to capture both the maximum and minimum values of a detector signal. This is accomplished by AC coupling the incoming detector signal so that the original minimum value shows up as a negative peak.

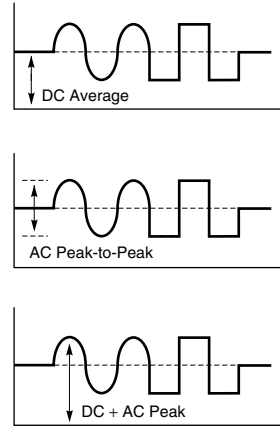


Figure 14 – Time Varying Signal Measurements - Many different measurements can be made on different portions of a time varying signal. The most common are: DC power, peak power, and peak-to-peak power.

The Model 2835-C can only measure periodic AC peak-to-peak signals within the frequency range: 50 - 7,000 Hz. Above 7,000 Hz, bandwidth limitations significantly effect the accuracy of the peak-to-peak measurement. Below 50 Hz, the AC coupling attenuates signals and measurement accuracy. However, low frequency measurements can be quantified in DC CONT measurement mode, see note below.

NOTE

Peak-to-peak measurements of frequencies below 50 Hz can be accomplished by observing the MAX-MIN value of the STATS buffer in DC CONT measurement mode, Sections 2.3.13 and 2.3.15. To do so, disable the analog filter, adjust the sample rate, Section 2.4.7 and the STATS buffer depth, Section 2.3.15, so that the actual maximum and minimum values will be captured by the stream of DC measurements.

3.8 Integration of Detector Signals

The Model 2835-C provides for making measurements that integrate incoming power detector signals to obtain an energy via the INTG mode, Section 2.3.13. In INTG mode, the display units indicate Joules since energy is the time integral of power:

$$E(\lambda) = \int_{t_0}^{t_1} P(\lambda) dt = \int_{t_0}^{t_1} \frac{S(t)}{R(\lambda)} dt$$

As the detector signal actually consists of a stream of digitized values, the integral becomes a numerical approximation using the trapezoid method, Figure 15. In order to maintain the highest accuracy for this numerical process, the sample rate is set at its maximum value of 400 Hz.

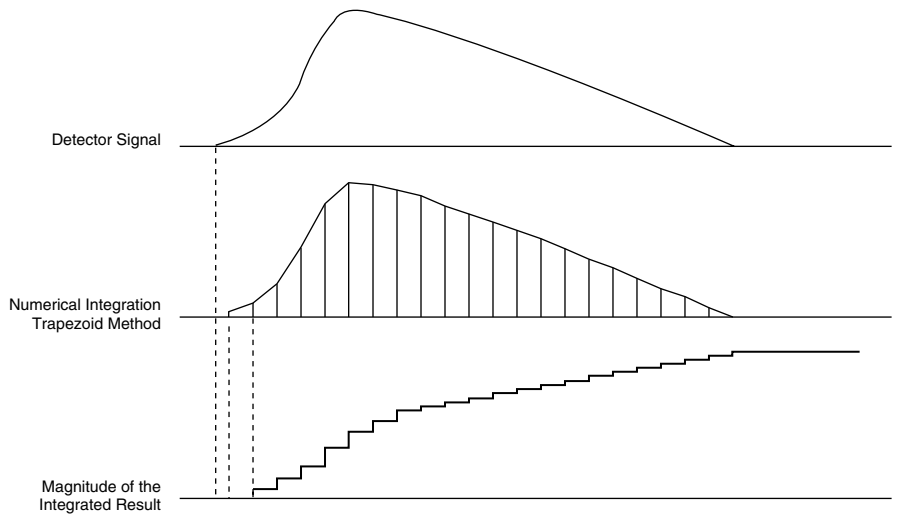


Figure 15 – Integrated Energy Via a Trapezoid Approximation - The INTG measurement mode performs a discrete integration at a 400 Hz sample rate.

Two common applications are natural extensions of the INTG measurement mode:

1. Pulse laser energy measurement using with a thermopile detector, Section 3.5
2. Energy from exposure over a period of time (dosage), Section 3.6.

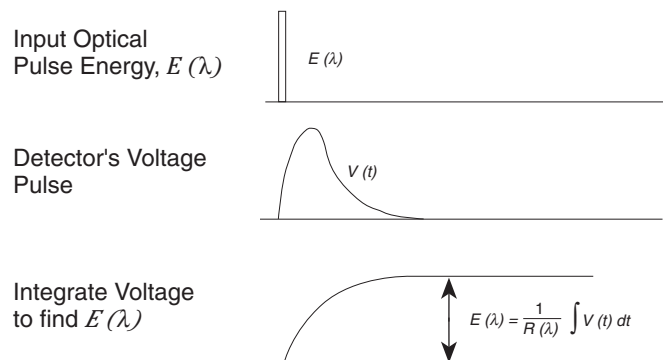


Figure 16. Measuring Laser Pulse Energy with a Thermopile - Thermopiles are often used to measure pulsed laser energy by integrating the response of the detector to the pulse.

3.9 Analog Output

The Model 2835-C provides a 0 to 2.5 volt, into 50W, BNC analog output for signal monitoring. The analog output is the actual amplified detector signal and is uncorrected for the effects of the detector's responsivity, AUTO CAL, and ZERO.

3.10 Measurement Considerations

This section describes detector characteristics, optical and electrical considerations, and environmental influences on optical measurements. In general, measurement accuracy is limited by the accuracy of the detector calibration. Accurate measurements, however, are also dependent upon proper set-up, control of temperature and illumination conditions and understanding the factors that affect optical measurements.

3.10.1 Detector Calibration and Accuracy

Newport Corporation calibrates its detectors using secondary standards directly traceable to the United States National Institute of Science and Technology (NIST) or to Great Britain's National Physical Laboratory (NPL). The details and accuracy of the calibration procedure vary with each detector model but a detailed description of the calibration results is supplied with each individually calibrated detector. In general, detector calibration accuracy varies from 2% to 5% in absolute terms and varies with wavelength. Each detector will also have some variation in response over its surface. Therefore, for the most reproducible measurements, light should illuminate the detector as uniformly as possible over as large an area as practical.

CAUTION

Avoid focusing a light source onto the detector surface. Inaccurate readings and possible detector damage may result. Consult the detector manual for saturation or damage thresholds.

NIST traceability requires that detectors be recalibrated on one year intervals. As individual detector responses change with time, especially in the ultraviolet, recalibration is necessary to assure confidence in the accuracy of the measurement. For the most reproducible measurements, the same detector should always be used for measurements which are to be directly compared.

3.10.2 Quantum Detector Temperature Effects

Semiconductor (Newport Low-Power) detectors, are affected by temperature. At long wavelengths, quantum detectors typically lose sensitivity with increasing temperature. Additionally, detector dark current increases exponentially with temperature.

Observed dark current is often dominated by the interaction between the detector and a meter's amplifier and is typically larger than the theoretical dark current limit. Silicon detectors are inherently quieter than germanium detectors due to their higher internal resistance and lower capacitance. The noise or drift in the dark current sets a lower bound on the measurement resolution which can be achieved with any given detector. Cooling a detector significantly lowers its dark current and dark current noise.

The observed dark currents can also be zeroed at any moment via the ZERO function. Since dark currents drift with temperature, the ZERO should be adjusted just prior to taking any measurements. If the detector temperature is constant, sensitivity changes and dark current drifts are significantly reduced.

3.10.3 Thermopile Detector Temperature Effects

Thermopile (Newport High-Power) detectors are significantly affected by temperature fluctuations arising from air flow disturbances. As the detector is a temperature measuring device, air flow disturbances set a practical lower limit on the power that a detector can measure. In order to get the most out of any thermopile detector, be careful to shield the detector from air flow disturbances. Common sources of disturbance are: air conditioners and people walking past.

3.10.4 Energy Detector Temperature Effects

Pyroelectric (Newport Energy) detectors are AC coupled devices and thus are not susceptible to temperature induced DC signal offsets or noise floor changes. One generally does not need to take much precaution with pyroelectric detectors except to make sure that their damage threshold is not exceeded.

3.10.5 Ambient and Stray Light

Ambient and stray light striking the detector should be considered when making a measurement. Ambient light can be distinguished from dark current (or the detector/meter noise floor) by either turning off or blocking the source and covering the detector face with opaque material such as a piece of black rubber.

Using the human hand to cover the detector is not advised because it emits a significant amount of infrared radiation and radiates a temperature significantly different from ambient. With the detector covered, a reading of the dark current may be made. Next, remove the material which is covering the detector and take another reading. The difference is the ambient light level.

NOTE

Changes in ambient light levels can occur from such factors as turning room lights on or off, or by moving people or equipment. Remember, if you can see your detector element, then your detector can see the light bouncing off you.

The effects of ambient light are greatly reduced when using a fiber-connectorized signal input to the detector. If free-space beam measurements are desired, using an attenuator will often improve the signal to ambient signal noise level. Wavelength-specific filters, such as optical cutoff, bandpass, or spike filters can be used if the signal wavelength spectrum permits. Other techniques to reduce stray light include using apertures, placing the detector in a box or other housing to shield the surface from light (or air currents) and turning off room and other polluting light sources.

3.10.6 Common Measurement Errors

The most common sources of optical measurement error are listed in Table 23 below. Other common errors are discussed in the preceding subsections of Section 3.10, Measurement Considerations.

Table 23 – Common Measurement Errors

Type of Error	What should be done?
Radiometry the detector.	Check that all of the light is actually hitting
Ambient Light before the measurement was made.	Check that any ambient light was ZEROed
Wavelength Calibration	Check that the Calibration Factor for the measurement wavelength is properly set.
Detector Saturation or Damage threshold.	Check that the optical power density remains below the detector's saturation or damage

NOTE

The Model 2835-C will indicate when a detector is being operated above its saturation or damage threshold by displaying “SA” in the measurement display area. Be aware that detectors often experience local saturation or damage without ever exceeding an overall saturation or damage threshold.

This only applies to semiconductor photodiodes and thermopiles used above maximum intermittent power.

Detector damage can still arise even when “SA” is not displayed. Be sure to follow your detector manual's user guidelines.

Section 4

Computer Interfacing

4.1 General Guidelines

The 2835-C has two computer interface ports: GPIB and RS-232C. The GPIB port conforms with the IEEE Standard 488.1 hardware standard and the IEEE 488.2 software standard. The protocol for the RS-232C port conforms as much as possible to the IEEE 488.2 software standard.

As specified by the IEEE 488.2 standard, there are two command types; device independent commands and device dependent commands. The device independent commands have the same meaning to all devices and are defined as part of the IEEE 488.2 standard. All device independent commands start with an asterisk (*). Device dependent commands do not start with an asterisk and have meanings unique to the Model 2835-C. A query is a command that invokes a response from the meter. All queries are terminated by a question mark (?). It is recommended that when a query is made, that the response to that query be read before other commands are issued. When a query is made to the GPIB interface, the MAV bit in the status byte should be checked by means of a serial poll to make sure that the data is available before reading the response. (See Appendix C.4 Status Byte)

4.2 Computer Interface Terminology

Listed below are key abbreviations and concepts used in the command reference section of this manual.

<...> Delimiting Punctuation

For the purposes of this manual, any string enclosed by <...> is considered to be a command, a string or a numerical argument. The punctuation <...> is used to symbolize the typographic limits of the command, string or argument in question.

<EOI> End or Identify

An IEEE 488.1 signal sometimes sent with the end-of-string character.

<IST> Individual Status

This status is generated by the status byte and parallel poll enable register. It is used in responding to parallel polls.

<CR> Carriage Return

The ASCII encoded byte 13 in decimal. (0D hex)

<LF> Line Feed

The ASCII encoded byte 10 in decimal. (0A hex)

<NL> New Line

Defined in the IEEE 488.2 standard as the ASCII encoded byte 10 in decimal. (0A hex)

<END> End of Message

Indicates the concurrent transmission of <EOI> with the a <NL> data byte.

(;) Semicolons

Used to separate commands within a single transmission (concatenation).

<number> Numerical Types

Numerical parameters are passed and returned as the actual ASCII characters in the string representation of the number. See the appendix for a detailed description of <number>.

<string> String Types

See the appendix for a detailed description of <string>

RS-232C Command Termination

When a command is received from the RS-232C port, either a <CR> or a <LF> is treated as the command terminator.

GPIB Command Termination

When a command is received from the GPIB port, either an <EOI> with a data byte, <NL>, or <CR> is treated as the command terminator. The recommended form of termination is an <EOI> sent with a <NL> denoted as <END> in this manual.

RS-232C Response Termination

In RS-232C echo mode responses are terminated by a <CR><LF> sequence. In RS-232C normal mode the <NL> character terminates the response.

GPIB Response Termination

All responses from the 2835-C GPIB port are terminated by an <END> sequence. Which is the concurrent transmission of <EOI> with a <NL> data byte.

4.3 Entering Remote Computer Interface Mode

When a command or query is received by the GPIB or RS-232 interface ports, the 2835-C automatically goes into remote interface mode. The message area on the display will indicate which remote port received the command.

While in remote mode, the keypad is disabled except for the power and <(LOCAL)> key. To get out of remote mode press the <(LOCAL)> key. If the 2835-C is in local lockout then the LLO remote command must be used to disable local lockout before you will be able to return to keypad control. The message area of the display will indicate when the keypad is in local lock out state.

4.4 RS-232C Communication

Before communicating with the 2835-C through the RS-232 port, proper cable connections must be made. Figure 17 shows the cable connections for communicating with the RS-232C port on the 2835-C.

Once cable connections are made, the baud rate and echo mode need to be set. Valid baud rates are 19.2K, 9600, 4800, 2400, and 1200. The parity, data bits, and stop bits are fixed at no parity, 8 data bits, and 1 stop bit.

RS-232C Parameters

Baud Rate	19.2K, 9600, 4800, 2400, or 1200
Parity	none
Data bits	8
Stop bits	1

When the echo mode is enabled the 2835-C generates a '>' prompt for every new line and all characters sent to the 2835-C are echoed back over the interface. Error messages are reported over the interface immediately. As the user is entering commands the line may be edited by using the backspace key (sending an ASCII decimal 08 code) or by using the DEL key (sending an ASCII decimal 127 code).

In echo mode the RS-232C port is interactive and especially useful when a dumb terminal type of device is used to communicate with the 2835-C.

When echo mode is disabled (normal mode) the 2835-C does not generate a prompt or echo characters back over the interface. Error messages must be accessed by using the *ERR? query. This is the default state of the echo mode.

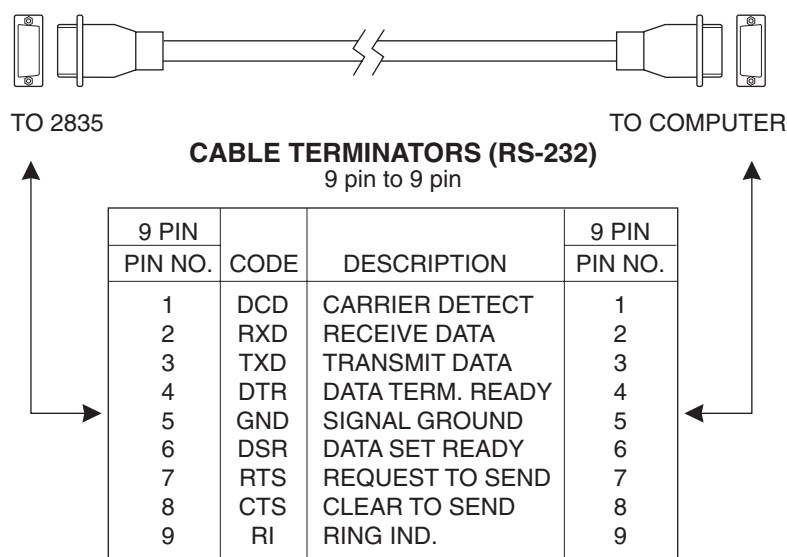
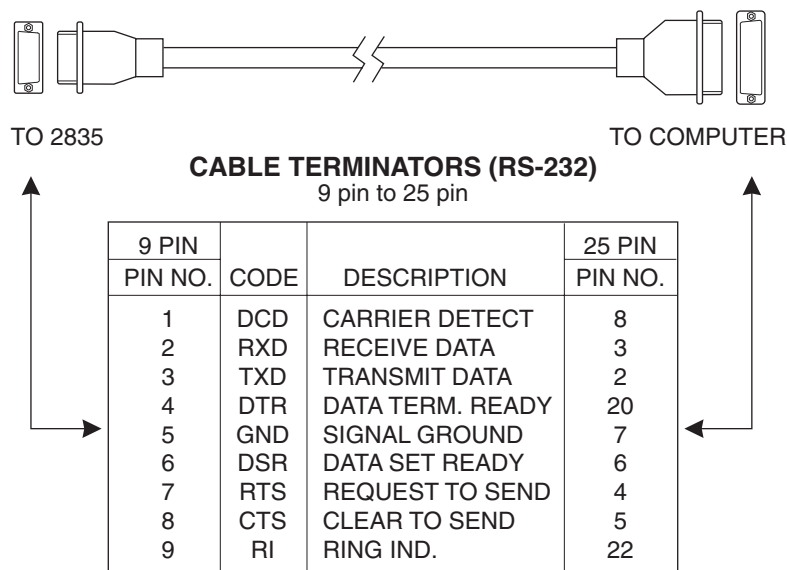


Figure 17–RS-232 Cable Connections

4.4.1 Setting Baud Rate and Echo Mode From The Keypad

The baud rate and echo mode are set from within the menu structure. To set the baud rate select the BAUD RATE option from the REMOTE SETUP menu. To set the echo mode select the ECHO MODE_n option from the REMOTE SETUP menu. (See Section 2.4.13)

4.4.2 Setting Baud Rate and Echo Mode From A Remote Interface

The baud rate cannot be set from the remote interface. To set the echo mode use the ECHO command. Use the ECHO? to see if the echo mode is enabled or disabled. (See Section 2.4.12)

4.5 RS-232C XON/XOFF Handshaking Protocol

The 2835-C uses the XON/XOFF handshaking protocol. When the 2835-C's RS-232 input buffer is nearly full the 2835-C will send an XOFF (ASCII encoded 19 decimal) character to the remote computer. The XOFF character informs the remote computer that it needs to suspend transmission until it receives the XON character from the 2835-C. When the buffer empties sufficiently the 2835-C will send an XON (ASCII encoded 17 decimal) to signal the remote computer that it can resume transmission.

The 2835-C will also respond to XOFF and XON characters sent to it from the remote computer. When the 2835-C receives an XOFF character it will suspend all transmission of characters to the remote computer until the XON character is received.

4.6 GPIB Communication

A variety of third party GPIB communication hardware and software, such as plug-in GPIB computer boards and Lab View software from National Instruments exists. The 2835-C should work with any of these as long as they adhere to the IEEE 488.1 standard. This manual assumes the user is familiar with one of these third party hardware/software packages. The table below summarizes the IEEE 488.1 capabilities supported by the 2835-C.

Table 24 – Model 2835-C IEEE 488.1 Capabilities Summary

Description	Command	Level of Capability
Source Handshake	SH1	complete capability
Acceptor Handshake	AH1	complete capability
Talker	T6	basic talker, serial poll, unaddress if MLA
Extended Talker	TE0	no capability
Listener	L4	basic listener, unaddress if MTA
Extended Listener	LE0	no capability
Service Request	SR1	complete capability
Remote Local	RL0	*no capability
Parallel Poll	PP1	remote configuration
Device Clear	DC1	complete capability
Device Trigger	DT0	no capability
Controller	C0	no capability

NOTE

* In order to better handle the RS-232C interface the 2835-C automatically goes into remote mode when a command is received from either of the remote computer interfaces. The LLO command is used to enable or disable the local lockout state.

4.6.1 Setting The GPIB Address

Before communicating with the 2835-C GPIB port the 2835-C GPIB address must be set. The address can be between 0 and 30. The address can only be set from the keypad and is accessed through the menu structure. To set the GPIB address select the REMOTE SETUP->GPIB ADDRESS option in the menu. The Model 2835-C GPIB is factory set to 5. (See Section 2.4.13)

Section 5

Command Reference

5.1 Model 2835-C Remote Interface Commands

The Model 2835-C remote commands can be divided into two groupings:

- i. Device Independent Commands, Section 5.2
- ii. Device Dependent Commands, Section 5.3

Table 25 lists the Device Independent Commands supported by the 2835-C while Table 26 lists the Device Dependent Commands supported the 2835-C. Following these tables are the detailed presentations of each command and its syntax and an example of its use.

Table 25 – Device Independent Status Commands

Command	Command Name
*CAL?	Calibration Query
*CLS	Clear Status
*ERR?	Error Query
*ESE	Event Status Enable
*ESE?	Event Status Enable Query
*ESR?	Event Status Register Query
*IDN?	Identification Query
*IST?	Individual Status Query
*OPC	Signal When Operation Complete
*OPC?	Operation Complete Query
*PRE	Parallel Poll Enable
*PRE?	Parallel Poll Enable Query
*RCL	Recall Device Configuration
*RST	Software Reset
*SAV	Save Device Configuration
*SRE	Service Request Enable
*SRE?	Service Request Enable Query
*STB?	Status Byte
*TST?	System Test
*WAI	Wait To Continue

Table 26 – Device Dependent Commands

Command	Command Name
ATTN_n	Attenuator Calibration Data Enable
ATTN_n?	Attenuator Calibration Data Enable Query
ATTNSN_n?	Attenuator Serial Number Query
AUTO_n	Auto Ranging Enable
AUTO_n?	Auto Ranging Enable Query
BARGRAPH	Bar Graph Enable
BARGRAPH?	Bar Graph Enable Query
CALDATE_n?	Calibration Date Query
CH?	Available Channel Query
DETMODEL_n?	Detector Mode Query
DETSN_n?	Detector Serial Number Query
DETSW_n	Detector Switch Select
DETSW_n?	Detector Switch Query
DISP	Display Brightness Select
DISP?	Display Brightness Query
DISPCH	Display Channel Select
DISPCH?	Display Channel Query
DS_n?	Data Store Value Query
DSBUF_n	Data Store Buffer Behavior Select
DSBUF_n?	Data Store Buffer Behavior Query
DSCLR_n	Data Store Clear
DSCNT_n?	Data Store Value Count Query
DSE_n	Data Store Enable
DSE_n?	Data Store Enable Query
DSSIZE_n	Data Store Buffer Size Select
DSSIZE_n?	Data Store Buffer Size Query
DSUNITS_n?	Data Store Units Query
ECHO	RS-232 Echo Mode Enable
ECHO?	RS-232 Echo Mode Enable Query
EVENT?	Device Event Register Query
EVENTEN	Device Event Enable Register
EVENTEN?	Device Event Enable Register Query
EXT	External Trigger Enable
EXT?	External Trigger Enable Query
EXTEDGE	External Trigger Edge Select
EXTEDGE?	External Trigger Edge Query
FILTER_n	Filter Select
FILTER_n?	Filter Query
LAMBDA_n	Lambda Select
LAMBDA_n?	Lambda Query
LLO	Local Lockout Enable
LLO?	Local Lockout Enable Query
MODE_n	Acquisition Mode Select
MODE_n?	Acquisition Mode Query
R?	Read Both Channel Measurements Query
R_n?	Read Single Channel Measurement Query
RANGE_n	Manual Range Level Select
RANGE_n?	Manual Range Level Query
REFSEL_n	Reference Source Select
REFSEL_n?	Reference Source Query
RESP_n?	Responsivity Query
RUN	Start Both Channel Acquisition
RUN_n	Start Single Channel Acquisition
RWS?	Read Both Channel Measurements with Status Query
RWS_n?	Read Single Measurement with Status Query

SFREQ	DC Sample Frequency Select
SFREQ?	DC Sample Frequency Query
SPREC	DC Sample Type Select
SPREC?	DC Sample Type Query
STMAX_n?	Statistics Buffer Maximum Value Query
STMEAN_n?	Statistics Buffer Mean Value Query
STMIN_n?	Statistics Buffer Minimum Value Query
STMXMN_n?	Statistics Buffer Maximum-Minimum Value Query
STSDEV_n?	Statistics Buffer Standard Deviation Query
STSIZE_n	Statistics Buffer Size Select
STSIZE_n?	Statistics Buffer Size Query
STOP	Stop Both Channel Acquisition
STOP_n	Stop Single Channel Acquisition
STOREF_n	Store Reference Reading
STOREF_n?	Store Reference Query Reading
STOZERO_n	Store Zero Signal
TONE	Tone Enable
TONE?	Tone Enable Query
TRIGOUT	Trigger Out Mode Select
TRIGOUT?	Trigger Out Mode Query
TRIGOUTFREQ	Trigger Out Frequency Select
TRIGOUTFREQ?	Trigger Out Frequency Query
TRIGOUTINTG	Trigger Out Integration Level Select
TRIGOUTINTG?	Trigger Out Integration Level Query
TRIGOUTLVL	Trigger Out Level Select
TRIGOUTLVL?	Trigger Out Level Query
TRIGOUTPOL	Trigger Out Polarity Select
TRIGOUTPOL?	Trigger Out Polarity Query
UNITS_n	Units Select
UNITS_n?	Units Query
USRCAL_n	User Calibration Enable
USRCAL_n?	User Calibration Enable Query
USRREF_n	User Defined Reference Value Select
USRREF_n?	User Defined Reference Value Query
USRRESP_n	User Responsivity Factor Select
USRRESP_n?	User Responsivity Factor Query
ZERO_n	Signal Zeroing Enable
ZERO_n?	Signal Zeroing Enable Query
ZEROVAL_n?	Zero Signal Query

NOTE:

n stands for A (channel A) or B (channel B).

5.2 Device Independent Commands

***CAL?**

Calibration Query

Syntax: *CAL?

Parameters: None

Function:

This routine performs an auto calibration sequence on both channels. The calibration sequence measures the amplifier offsets at all gain settings. These measurements are stored and subtracted from subsequent measurements. Acquisition is suspended during auto calibration and no other remote commands will be processed until the auto calibration sequence is complete.

Returns: <number>

<number> represents the integer 0. This return value is generated when autocalibration is complete.

Related Commands: None

***CLS**
Clear Status

Syntax: *CLS

Parameters: None

Function:

This command clears the Standard Event Status register, device event register, and the error queue. As a result of this command the status byte is cleared except for the message available bit (bit 4). Because this command does not clear the input and output buffers the MAV bit will not be affected. The *CLS command also cancels any outstanding *OPC and OPC? commands.

Related Commands: *ESR?, *EVENT?, *IST?, *STB?

***ERR?**
Error Query

Syntax: *ERR?

Parameters: None

Function:

Returns (and removes) oldest error message from the error queue. The error message is made up of an error code and a text description of the error (see the Appendix B). A maximum of 10 errors can be stored in the queue. If no errors are stored in the queue, a message to that effect will be returned. Errors that occur due to commands and queries sent from the RS-232 interface will be stored in the queue only if the echo mode is disabled. When the echo mode is enabled the error messages are automatically returned to the user when the error occurs. See Appendix B for a description of the error messages. The command *CLS will clear the error queue.

Returns: <errno>,<description>

<errno> is of type <number> and represents a negative integer error code. <description> is of type <string> in double quote format, and describes the error. See the appendix for a list of possible error messages.

Related Commands: *CLS, *STB?

***ESE**

Event Status Enable

Syntax: *ESE <mask>

Parameters:

<mask> is an integer in the range 0 to 255 inclusive.

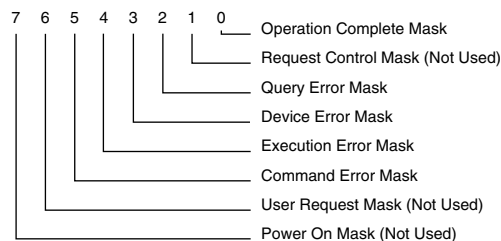
<mask> is written to the Event Status Enable register.

Function:

The Event Status Enable register is AND'd with the Event Status register. If any bit is set in the result of this AND'ing operation the ESB bit (bit 5) of the Status Byte will be set. The Status Byte can be used in conjunction with the Service Request Enable register to generate a service request.

The Event Status Enable register is an 8 bit, bit mapped register. Any bit set to 1 in the Event Status Enable Register allows the corresponding bit in the Event Status Register to set the ESB bit (bit 5) in the Status Byte. Any bit set to 0 disables the corresponding bit in the Event Status Register from setting the ESB bit. The Event Status Enable register is set to 0 upon power-up. See Appendix C for a detailed description of the Event Status Register.

The significance of each bit in the Event Status Enable register is shown below:



Event Status Enable Register

Related Commands: *CLS, *ESE?, *ESR?, EVENT?,

EVENTEN, EVENTEN?, *IST?, *PRE, *PRE?, *SRE, *SRE?, *STB?

***ESE?**

Event Status Enable Query

Syntax: *ESE?

Parameters: None

Function:

This query returns the contents of the Event Status Enable register. See the *ESE command for a description of the Event Status Enable register. See Appendix C for a detailed description of the Event Status Register.

Returns: <mask>

<mask> is of type <number> and represents an unsigned integer in the range 0 to 255.

Related Commands: *CLS, *ESE, *ESR?, EVENT?, EVENTEN, EVENTEN?, *IST?, *PRE, *PRE?, *SRE, *SRE?, *STB?

***ESR?**

Event Status Register Query

Syntax: *ESR?

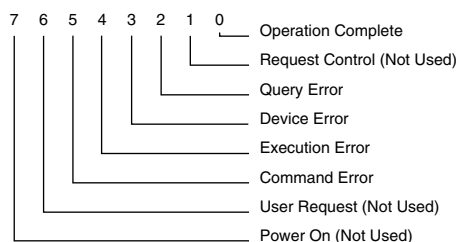
Parameters: None

Function:

This query returns the contents of the Event Status Register and clears the Event Status Register except for bit 3. Bit 3 is determined by the state of the Device Event and Device Event Enable registers. The *CLS command will also set this register to 0.

The Event Status register is AND'ed with the Event Status Enable register. If any bit is set in the result of this AND'ing operation the ESB bit (bit 5) of the Status Byte will be set. The Status Byte can be used in conjunction with the Service Request Enable register to generate a service request. See Appendix C for a detailed explanation of the Event Status Register.

The Event Status Register is an 8 bit, bit mapped register, with each bit signifying a different condition. The bits are listed below, most significant bit first:



Standard Event Status Register

Returns: <value>

<value> is of type <number> and represents an unsigned integer in the range 0 to 255.

Related Commands: *CLS, *ESE, *ESE?, EVENT, EVENTEN, EVENTEN?, *IST?, *PRE, *PRE?, *SRE, *SRE?, *STB?

***IDN?**
Identification Query

Syntax: *IDN?

Parameters: None

Function:

This query causes the 2835-C to return device identification information. The information returned is manufacturer, model, serial number, and firmware version.

Returns: <manufacturer>,<model>,<serial>,<version>

<manufacturer> is of type <string> using the no quotes format. The 2835-C always returns Newport Corp.

<model> is of type <string> using the no quotes format.

<serial> is of type <string> using the no quotes format. This is always a '0'.

<version> is of type <string> using the no quotes format. This text identifies the internal firmware version of the 2835-C. This text consists of two sections, separated by two underscores. The first section is the revision level, and the second is the revision date.

Related Commands: None

***IST?**
Individual Status Query

Syntax: *IST?

Parameters: None

Function:

This query returns the current state of the IEEE 488.1

<IST> local message. The <IST> message is generated by the parallel poll status system. If any bit is set in the Status Byte and it's corresponding bit is set in the Parallel Poll Enable register, then the <IST> message is set true (a value of 1). Otherwise the <IST> message is set false (a value of 0).

When a parallel poll is conducted with the 2835-C configured to respond to it, the <IST> message is compared to the <S BIT> (sense bit). If they are the same then the configured data line will be driven true in response to the parallel poll.

Both the <S BIT> and the data line driven during a parallel poll can be selected by the IEEE 488.1 parallel poll remote configuration command.

Returns: <sbit>

<sbit> is of type <number> which represents the integer 0 or 1.

Related Commands: *PRE, *PRE?, *STB?

***OPC** **Signal When Operation Complete**

Syntax: *OPC

Parameters: None

Function:

This command provides a means of synchronizing operations between the 2835-C and the host. This command sets the Operation Complete bit (bit 0) of the Event Status register to 1 when all pending operations have completed. This bit will remain 1 until cleared by the *CLS command or the *ESR? query.

This command can be used in conjunction with the *ESE and *SRE commands to cause a Service Request on the GPIB bus when all pending operations are complete. Once the *OPC command is received, it will not set the operation complete bit to 1 while any of the following operations are in progress:

Single acquisition in progress

Data store enabled with fixed data store buffer

The meter is ranging

Related Commands: *ESE, *ESE?, *ESR?, *OPC?, *SRE, *SRE?, *STB?, *WAI

***OPC?** **Operation Complete Query**

Syntax: *OPC?

Parameters: None

Function:

This query provides a means of synchronizing operations between the 2835-C and the host. This command generates a response when all pending operations have completed.

When all pending operations are complete, this query will cause the Message Available bit (bit 4) of the Status Byte to be set because of the response generated. By enabling bit 4 of the Service Request Enable register (see *SRE), the *OPC? query can be used to cause a Service Request on the GPIB bus when all pending operations are complete.

Once the *OPC? query is received, the response will not be generated while any of the following operations are in progress:

Single acquisition in progress

Data store enabled with fixed data store buffer

The meter is ranging

The *CLS or *RST commands, as well as a GPIB interface device clear, will cancel the operation complete query.

Returns: 1

The response is generated when all pending operations are complete.

Related Commands: *OPC, *SRE, *SRE?, *STB?, *WAI

***PRE**
Parallel Poll Enable

Syntax: *PRE <mask>

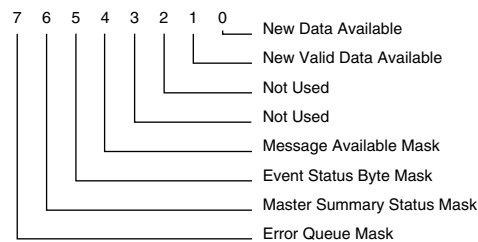
Parameters:

<mask> is an integer in the range 0 to 255 inclusive. The value of this number is written into the Parallel Poll Enable register.

Function:

This command sets the Parallel Poll Enable register bits. The Parallel Poll Enable register is used in conjunction with the Status Byte to generate the IST (Individual Status) message. See Appendix C for a detailed description of the Event Status Register.

The Parallel Poll Enable Register is set to 0 at power-on. The Parallel Poll Enable Register is an 8-bit, bit mapped register, with each bit signifying a different condition. The bits and their significance are detailed below:



Parallel Poll Enable Register

Related Commands: *IST, *PRE?, *STB?*PRE?

Parallel Poll Enable Query

Syntax: *PRE?

Parameters: None

Function:

This query returns the contents of the Parallel Poll Enable register. See the *PRE command for a description of the Parallel Poll Enable register. See Appendix C for a detailed description of the Event Status Register.

Returns: <mask>

<mask> is of type <number> and represents an unsigned integer in the range 0 to 255.

Related Commands: *IST, *PRE, *STB?

***RCL**

Recall Meter Configuration

Syntax: *RCL <config>

Parameters:

<config> is of type <number> which rounds to an integer in the range 0 to 9. This number specifies a configuration storage buffer. The number 0 stands for the default configuration buffer.

Function:

This command configures the meter to a configuration previously stored in non-volatile memory (see *SAV), or to the default configuration for the attached detector. This command will fail if the requested configuration is incompatible with the detector attached to the meter or empty. The parameters affected by this command are those listed in Table 9.

Returns: None

Related Commands: *RST, *SAV

***RST**

Reset

Syntax: *RST

Parameters: None

Function:

This sets the meter to the default configuration, see Table 9, for the detector attached to it. It also cancels any outstanding *OPC or *OPC? commands.

Related Commands: *RCL, *SAV

***SAV**

Save Meter Configuration

Syntax: *SAV <config>

Parameters:

<config> is of type <number> which rounds to an integer in the range 1 to 9. This number specifies a configuration storage buffer.

Function:

This command saves the configuration of the meter to the configuration specified by the parameter. The parameters saved by this command are those listed in Table 9.

Related Commands: *RCL, *RST

***SRE** **Service Request Enable**

Syntax: *SRE <mask>

Parameters:

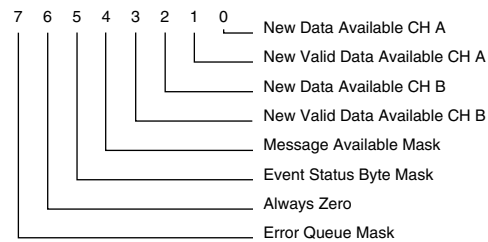
<mask> is an integer in the range 0 to 255 inclusive. The value of this number is written into the Service Request Enable register.

Function:

The Service Request Enabled Register is used in conjunction with the Status Byte to generate service requests on the GPIB bus. See Appendix C for a detailed description of the Event Status Register.

If a bit is set in the Service Request Enable register and it's corresponding bit is set in the Status Byte Register, then a service request will be generated once for the for the given event. When service is being requested, bit 6 is set in the byte returned by a serial poll of the 2835-C.

The Service Request Enable Register is an 8-bit, bit mapped register, with each bit signifying a different condition. The bits are listed below, most significant bit first:



Service Request Enable Register

Related Commands: *SRE?, *STB?

***SRE?** **Service Request Enable Query**

Syntax: *SRE?

Parameters: None

Function:

This query returns the contents of the Service Request Enable register. See the *SRE command for a description of the Service Request Enabled register. See Appendix C for a detailed description of the Event Status Register.

Returns: <mask>

<mask> is an integer in the range 0 to 255 inclusive, except that bit 6 is always a zero.

Related Commands: *SRE, *STB?

***STB?** **Status Byte Query**

Syntax: *STB?

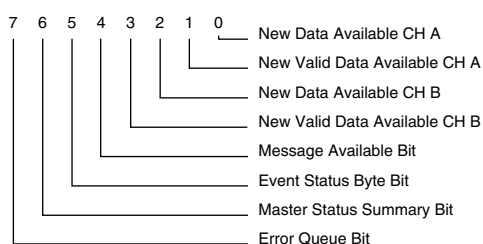
Parameters: None

Function:

This query returns the contents of the Status Byte which records current system conditions. Appendix C describes the Status Byte in detail.

If a bit is set in the Status Byte Register and it's corresponding bit is set in the Service Request Enable Register, then a service request will be generated once for the given event. When service is being requested, bit 6 is set in the byte returned by a serial poll of the 2835-C.

The Status Byte Register is an 8-bit, bit mapped register, with each bit signifying a different condition. When a bit is set, or has a value of one, then the condition is true. The bits are listed below, most significant bit first:



Status Byte Register.

Note that the *CLS common command clears most of the bits in the Status Byte.

Returns: <value>

<value> is an integer in the range 0 to 255 inclusive.

Related Commands: *CLS, *SRE, *SRE?

***TST?** **Self Test Query**

Syntax: *TST?

Parameters: None

Function:

As defined in the IEEE 488.2 standard, the *TST? command causes the device to return a result of a self-test, indicating whether or not the unit completed the self-test without any errors. The self test is only administered upon reset or power up. The query does not initiate the self test and only returns a 0 indicating that the system passed its earlier self test.

Returns: 0

Related Commands: None

***WAI**
Wait to continue

Syntax: *WAI

Parameters: None

Function:

The *WAI command causes the device to wait until all pending operations are complete before processing any commands waiting in the input queue. Care should be taken when using this command since it is possible to overflow the input queue by continuing to write commands to the 2835-C while the *WAI command is in effect.

Note that the *WAI command will essentially “lock out” the remote interfaces until all pending operations are complete. The *OPC command and *OPC? query differ from *WAI in this area, as they allow other commands to be processed while they are waiting for pending operations to complete.

Once the *WAI command is received, it will be in effect while any of the following operations are in progress:

- Single acquisition in progress

- Data store in progress with fixed data store buffer

- The meter is ranging

The *WAI command can only be canceled by a GPIB interface device clear or by turning the meter off and on.

Related Commands: *OPC, *OPC?

5.3 Device Dependent Commands

ATTN_n

Attenuator Calibration Data Enable

Syntax: ATTN_n <state>
where n is either A or B

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0, then detector responsivity data is used to calculate readings. If <state> is 1, then the detector + attenuator responsivity data is used to calculate readings.

Function:

This command enables or disables the use of attenuator responsivity data when user calibration is disabled. If enabled, responsivity data collected with the attenuator attached to the detector is used in data calculations. If disabled, responsivity data collected with the detector alone is used in data calculations.

This command will generate an error if the attached detector does not support an attenuator.

Related Commands: ATTN_n?, ATTNSN_n?, LAMBDA_n,
LAMBDA_n?, RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n,
USRRESP_n?

ATTN_n?

Attenuator Calibration Data Enable Query

Syntax: ATTN_n?
where n is either A or B

Parameters: None

Function:

This query returns a value showing whether or not attenuator responsivity data will be used when user calibration is disabled.

Returns: <state>

<state> is of type <number> which represents the integer 0 if detector responsivity data is being used or 1 if detector + attenuator responsivity data is being used.

Related Commands: ATTN_n, ATTNSN_n?, LAMBDA_n, LAMBDA_n?,
RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n, USRRESP_n?

ATTNSN_n? **Attenuator Serial Number Query**

Syntax: ATTNSN_n?
where n is either A or B

Parameters: None

Function:

This query returns the serial number of the attenuator that was calibrated with the attached detector.

When using an attenuator this serial number must match the serial number of the attenuator being used or data calculations will be inaccurate.

Returns: <attnSN>

<attnSN> is of type <string> in the double quote format.

If no attenuator serial number is available, the query returns an empty double quoted string, "".

Related Commands: CALDATE_n?, DETMODEL_n?, DETSN_n?

AUTO_n **Auto Ranging Enable**

Syntax: AUTO_n <state>
where n is either A or B

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0 then manual ranging is enabled. If <state> is 1 then auto ranging is enabled.

Function:

This command enables or disables auto ranging. If enabled and in a continuous acquisition mode, a signal range is automatically selected to suit the input signal. If disabled, the meter needs to be manually ranged.

Related Commands: AUTO_n?, RANGE_n, RANGE_n?

AUTO_n? **Auto Ranging Enable Query**

Syntax: AUTO_n?
where n is either A or B

Parameters: None

Function:

This query returns a value showing whether the auto ranging feature is enabled or disabled.

Returns: <state>

<state> is of type <number> which represents the integer 0 if auto ranging is disabled or 1 if auto ranging is enabled. If enabled, the meter will auto range when in continuous acquisition modes.

Related Commands: AUTO_n, RANGE_n, RANGE_n?

BARGRAPH

Bar Graph Enable

Syntax: BARGRAPH <state>

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0 then the bargraph is disabled. If <state> is 1 then the bargraph is enabled.

Function:

This command enables or disables the bargraph feature. If enabled, the bargraph will be displayed when the units of measurement are not ratio type of units and the meter is not in INTG mode.

Related Commands: BARGRAPH?, TONE, TONE?

BARGRAPH?

Bar Graph Enable Query

Syntax: BARGRAPH?

Parameters: None

Function:

This query returns a value showing whether the bargraph is enabled or disabled. Note: The bargraph is not displayed while in either GPIB or RS-232 mode regardless of enabled status.

Returns: <state>

<state> is of type <number> which represents the integer 0 if the bar graph is disabled and 1 if the bar graph is enabled.

Related Commands: BARGRAPH, TONE, TONE?

CALDATE_n?

Calibration Date Query

Syntax: CALDATE_n?
where n is either A or B

Parameters: None

Function:

This query returns the calibration date of the detector as read from the memory module attached to the 2835-C.

Returns: <model>

<model> is of type <string> in the double quote format.

Related Commands: ATTNSN_n?, DETMODEL_n?, DETSN_n?

CH?

Available Channel Query

Syntax: CH?

Parameters: None

Function:

This query is used to see which channels are available in the meter. A channel is available if a calibration module was plugged into that channel on powerup and passed the self test.

Returns: <channels>

<channels> is of type <string> in the double quote format. The possible values of <string> are:

“A” - only channel A is available

“B” - only channel B is available

“AB” - Both channel A and B are available

Related Commands: None

DETMODEL_n?

Detector Model Query

Syntax: DETMODEL_n?
where n is either A or B

Parameters: None

Function:

This query returns the model number of the detector that belongs to the memory module attached to the specified channel.

Returns: <model>

<model> is of type <string> in the double quote format.

Related Commands: ATTNSN_n?, CALDATE_n?, DETSN_n?

DETSN_n?

Detector Serial Number Query

Syntax: DETSN_n?
where n is either A or B

Parameter: None

Function:

This query returns the serial number of the detector that belongs to the memory module attached to the specified channel.

This serial number must match the serial number of the detector attached to the 2835-C.

Returns: <detSN>

<detSN> is of type <string> in the double quote format.

Related Commands: ATTNSN_n?, CALDATE_n?, DETMODEL_n?

DETSW_n

Detector Switch Setting

Syntax: DETSW_n <setting>
where n is either A or B

Parameter:

<setting> is of type <string> in a double quoted format. The possible values of <setting> are “S” for short, “I” for intermediate, and “L” for long.

Function:

The 818J-S10 and the 818J-09 have a switch setting for short, intermediate and long energy pulse measurement. This command must be used to inform the 2835-C what the switch is set to when using these detectors.

If the attached detector doesn’t support a switch then this command will cause an execution error.

Related Commands: DETSW_n?

DETSW_n?

Detector Switch Setting Query

Syntax: DETSW_n?
where n is either A or B

Parameter: None

Function:

This query returns the switch setting assumed by the 2835-C. This setting must match the actual switch setting on the detector or readings will be inaccurate.

Returns: <setting>

<setting> is of type <string> in the double quote format. The possible values of <setting> are “S” for short, “I” for intermediate, and “L” for long. If the attached detector doesn’t support a switch the return value will be “N/A”.

Related Commands: DETSW_n

DISP

Display Brightness

Syntax: DISP <level>

Parameters:

The parameter <level> is of type <number> which rounds to an integer in the range 0 to 3. 0 corresponds to off, 1 to dim, 2 to normal, and 3 to bright.

Function:

This command sets the brightness of the display to a desired level. The valid levels are off, dim, normal, and bright.

Returns: None

Related Commands: DISP?

DISP?
Display Brightness Query

Syntax: DISP?

Parameters: None

Function:

This query returns the brightness level of the display.

Returns: <level>

<level> is of type <number> where 0 corresponds to off, 1 to dim, 2 to normal, and 3 to bright.

Related Commands: DISP

DISPCH
Display Channel Select

Syntax: DISPCH <channel>

Parameters:

<channel> is of type <string> in a double quoted format.

The possible values of <channel> are “A” for channel A and “B” for channel B.

Function:

This command is used to select the display channel of the meter. If the requested channel is not available then this command will cause an error.

Returns: None

Related Commands: CH?, DISPCH?

DISPCH?
Display Channel Query

Syntax: DISPCH?

Parameters: None

Function:

This command is used to see which channel is currently selected as the display channel.

Returns: <channel>

<channel> is of type <string> in a double quoted format. The possible values of <channel> are “A” for channel A and “B” for channel B.

Related Commands: CH?, DISPCH

DS_n?

Data Store Value Query

Syntax: DS_n? <number>

where n is either A or B

Parameters:

The parameter <number> rounds to an integer and represents the data store value that is being queried. The range of <number> is 1 to n; where n is the number of values stored in the data store buffer (see DSCNT_n?). A parameter of 1 represents the oldest data value in the data store, 2 the second oldest and so on.

Function:

This query is used to query an individual data value from the data store buffer. If the parameter is out of range then an execution error message will be generated and nothing will be returned. It is recommended to use the DSCNT_n? query to determine how many values have been stored in the data buffer.

Returns: <status>,<number>

<status> is an integer that represents the status of the returned value. <status> will be 0 for ok, 1 for over range, 2 for detector saturated, 3 for data error and 4 for ranging.

<number> is a floating point number in the exponential format.

Related Commands: DSCLR_n, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUFF_n, DSBUFF_n?

DSBUF_n

Data Store Buffer Behavior Select

Syntax: DSBUFF_n <state>

where n is either A or B

Parameters:

The parameter <state> is of type <number>. If the number rounds to 0 then data store FIXED BUFFER behavior is enabled. Otherwise, data store SLIDE BUFFER behavior is enabled.

Function:

This command selects data store FIXED BUFFER or SLIDE BUFFER behavior.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUFF_n? DS_n

DSBUF_n?
Data Store Buffer Behavior Query

Syntax: DSBUF_n?

where n is either A or B

Parameters: None

Function:

This query returns a value showing whether data store FIXED BUFFER or SLIDE BUFFER behavior is enabled.

Returns: <state>

<state> is of type <number> and represents an integer 0, if FIXED BUFFER behavior is enabled, or 1, if data store SLIDE BUFFER behavior is enabled.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DS_n

DSCLR_n
Data Store Clear

Syntax: DSCLR_n

where n is either A or B

Parameters: None

Function:

This command is used to clear the data store buffer.

Related Commands: DS_n?, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DSBUF_n?

DSCNT_n?
Data Store Value Count Query

Syntax: DSCNT_n? <number>

where n is either A or B

Parameters: None

Function:

This query returns the number of data values stored in the data store buffer.

Returns: <count>

<count> is of type <number> and represents an integer in the range of 1 to the size of the data store buffer.

Related Commands: DS_n?, DSCLR_n, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DSBUF_n?

DSE_n **Data Store Enable**

Syntax: DSE_n <state>

where n is either A or B

Parameters:

The parameter <state> is a <number> which is either 0 or 1. If the number is 0, data storing is disabled. If the number is 1, data storing is enabled.

Function:

This command enables or disables data storing. Note that data storing cannot be enabled when the data store buffer is full and FIXED BUFFER behavior is enabled. The user must first either clear the data store buffer or enable SLIDE BUFFER behavior and then enable data storing.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n?, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DSBUF_n? DS_n

DSE_n? **Data Store Enable Query**

Syntax: DSE_n?

where n is either A or B

Parameters: None

Function:

This query returns a value showing whether or not data storing is enabled or disabled.

Returns: <state>

<state> is of type <number> and represents an integer 0, if data storing is disabled, or 1, if data storing is enabled.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n, DSSIZE_n, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DSBUF_n? DS_n

DSSIZE_n **Data Store Buffer Size Select**

Syntax: DSSIZE_n <size>

where n is either A or B

Parameters:

The parameter <size> is of type <number> which rounds to an integer in the range 1 to 1000. The parameter represents the size of the data buffer to be used for data storing.

Function:

This command sets the size of the buffer used for data storing. Note that the data buffer is cleared automatically when this command is used and all previously stored data will be gone.

Related Commands: DS_n?, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n?, DSUNITS_n?, DSBUF_n, DSBUF_n? DS_n,

DSSIZE_n?

Data Store Buffer Size Query

Syntax: DSSIZE_n?

where n is either A or B

Parameters: None

Function:

This query returns the data store buffer size.

Returns: <size>

<size> is of type <number> and represents an integer of the range 1 to 1000.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSUNITS_n?, DSBUF_n, DSBUF_n?, DS_n

DSUNITS_n?

Data Store Buffer Units Query

Syntax: DSUNITS_n?

where n is either A or B

Parameters: None

Function:

This query returns the units of the data stored in the data store buffer.

Returns: <units>

<units> is of type <string> in the double quote format.

The possible values of units are: "A", "V", "W", "W/CM", "dBm", "dB", "REL", "J", "J/CM", "Erg", or "E/CM" depending on the detector and acquisition mode.

Related Commands: DS_n?, DSCLR_n, DSCNT_n?, DSE_n, DSE_n?, DSSIZE_n, DSSIZE_n?, DSBUF_n, DSBUF_n?

ECHO

RS-232 Echo Mode Enable

Syntax: ECHO <enable>

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0 then RS-232 echo mode is disabled. If <state> is 1 then RS-232 echo mode is enabled.

Function:

This command enables or disables the echo mode for RS-232 communication.

Related Commands: ECHO?

ECHO? RS-232 Echo Mode Enable Query

Syntax: ECHO?

Parameters: None

Function:

This query returns a value showing whether the RS-232 echo mode is enabled or disabled.

Returns: <state>

<state> is of type <number> that represents the integer 0, if the echo mode is disabled, or 1, if the echo mode is enabled.

Related Commands: ECHO

EVENT? Device Event Register Query

Syntax: EVENT?

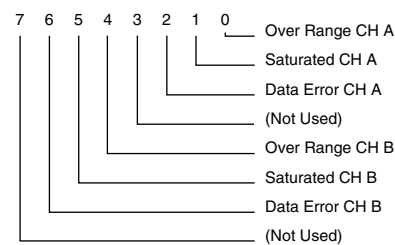
Parameters: None

Function:

This query returns the contents of the Device Event register and sets the Device Event register to 0. The *CLS command will also set this register to 0.

The Device Event register is AND'ed with the Device Event Enable register. If any bit is set in the result of this AND'ing operation then Device Error bit (bit 3) in the Standard Event Status register will be set.

The Device Event register is an 8 bit, bit mapped register, with each bit signifying a different condition. The bits are listed below, most significant bit first:



Device Event Register

Returns: <event>

<event> is an integer in the range from 0 to 255 inclusive.

Related Commands: *ESE, *ESE?, *ESR?, EVENTEN, EVENTEN?

EVENTEN

Device Event Enable Register

Syntax: EVENTEN <mask>

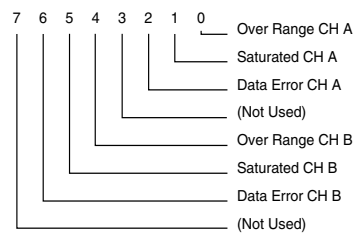
Parameters:

<mask> is an integer in the range 0 to 255 inclusive. The value of this number is written into the Device Event Enable register.

Function:

The Device Event Enable register is AND'ed with the Device Event register. If any bit is set in the result of this AND'ing operation then Device Error bit (bit 3) in the Standard Event Status register will be set.

The Device Event Enable register is an 8 bit, bit mapped register. Any bit set to 1 in the Device Event Enable register allows the corresponding bit in the Device Event register to set the Device Error bit (bit 3) in the Standard Event Status register. Any bit set to 0 disables the corresponding bit in the Device Event register from setting the Device Error bit.



Device Event Register

Related Commands: EVENT?, EVENTEN?, *ESE, *ESE?, *ESR?

EVENTEN?

Device Event Enable Register Query

Syntax: EVENTEN?

Parameters: None

Function:

This query returns the contents of the Device Event Enable register. See the EVENTEN command for a description of the Device Event Enable register.

Returns: <mask>

<mask> is of type <number> and represents an unsigned decimal integer in the range from 0 to 255.

Related Commands: EVENT?, EVENTEN, *ESE, *ESE?, *ESR?

EXT

External Trigger Enable

Syntax: EXT <enable>

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0 then the external trigger is disabled. If <state> is 1 then the external trigger is enabled.

Function:

This command enables or disables the external trigger input on the back panel.

Related Commands: EXT?, EXTEDGE, EXTEDGE?

EXT?

External Trigger Enable Query

Syntax: EXT?

Parameters: None

Function:

This query returns a value showing whether the external trigger input is enabled or disabled.

Returns: <state>

<state> is of type <number> that represents the integer 0, if the external trigger input is disabled, or 1, if the external trigger is enabled.

Related Commands: EXT, EXTEDGE, EXTEDGE?

EXTEDGE

External Trigger Edge Select

Syntax: EXTEDGE <edge>

Parameters:

The parameter <edge> is either 0 or 1. If <edge> is 0 then the external trigger is defined as a falling edge. If <edge> is 1 then the external trigger is defined as a rising edge.

Function:

This command defines whether the external trigger input on the back panel is falling edge or rising edge active.

Related Commands: EXT, EXT?, EXTEDGE

EXTEDGE?

External Trigger Edge Query

Syntax: EXTEDGE?

Parameters: None

Function:

This query returns a value showing which external trigger edge is the active edge.

Returns: <edge>

<edge> is of type <number> and represents 0, if the external trigger is defined as a falling edge, or 1, if the external trigger is defined as a rising edge.

Related Commands: EXT, EXT?, EXTEDGE

FILTER_n **FilterSelect**

Syntax: FILTER_n <filter>

where n is either A or B

Parameters:

The parameter <filter> is an integer in the range 0 to 3 inclusive. 0 corresponds to no filtering, 1 to analog filter only, 2 to digital averaging filter only, and 3 to analog and digital filters combined.

Function:

This command selects a filter combination to be used to condition readings taken from the detector. The choices are to have no filter at all, an analog filter, digital averaging filter, and both the analog and digital filters. The digital filter processes every signal reading by averaging it with the last 9 readings taken.

All detectors signals are filtered to some extent even without the analog filter feature. By enabling the analog filter the signal is filtered by a 5 Hz low pass instead of the standard filter. The analog filter feature only applies to Low Power detectors when not in P-P mode. For High Power and Energy detectors, the analog filter feature is not available.

If an attempt is made to enable the analog filter for a detector that does not support it, an execution error will occur.

Related Commands: FILTER_n?

FILTER_n? **Filter Query**

Syntax: FILTER_n?

where n is either A or B

Parameters: None

Function:

This query returns a value that represents the filter combination currently being used to condition readings.

Returns: <filter>

<filter> is of type <number> in a range of 0 to 3 with 0 corresponding to no filter, 1 to analog filter only, 2 to digital averaging filter only, and 3 to analog and digital filters combined.

This query will always return a 0 or 2 for High Power and Energy detectors because the use of the analog filter is not allowed with these types of detectors.

Related Commands: FILTER_n

LAMBDA_n

Lambda Select

Syntax: LAMBDA_n <wavelength>

where n is either A or B

Parameters:

The parameter <wavelength> is of type <number> which rounds to an integer with a range that depends on the detector being used. The units of <wavelength> are nanometers and should correspond to the wavelength of the light source being measured.

Function:

This command is used to specify the wavelength of light being measured. This will insure that the proper responsivity calibration point will be used when the 2835-C calculates measurement values.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n?, RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n, USRRESP_n?

LAMBDA_n?

Lambda Query

Syntax: LAMBDA_n?

where n is either A or B

Parameters: None

Function:

This query returns the wavelength that corresponds to the responsivity calibration point currently being used by the 2835-C in measurement calculation.

Returns: <wavelength>

<wavelength> is of type <number> and represents an integer. The units of <wavelength> are nanometers.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n, USRRESP_n?

LLO

Local Lockout Enable

Syntax: LLO <state>

Parameters:

The parameter <state> is either 0 or 1. If <state> is 0 then local lock-out is disabled. If <state> is 1 then local lock-out is enabled.

Function:

This command enables or disables the local lockout.

Returns: None

Related Commands: LLO?

LLO?
Local Lockout Enable Query

Syntax: LLO?

Parameters: None

Function:

The query returns the state of the local lockout.

Returns: <state>

<state> is of type <number> and represents the integer 0, if local lockout is disabled, or 1, if local lockout is enabled.

Related Commands: LLO

MODE_n
Acquisition Mode Select

Syntax: MODE_n <mode>

where n is either A or B

Parameters:

The parameter <mode> is of type <string>. Its range depends on the detector attached to the meter. The valid values for this parameter are listed below.

"DCSNGL"	specifies DC single mode
"DCCONT"	specifies DC continuous mode
"INTG"	specifies integration mode
"PPSNGL"	specifies P-P single mode
"PPCONT"	specifies P-P continuous mode
"SNGLPULSE"	specifies single pulse mode
"CONTPULSE"	specifies continuous pulse mode

Function:

This command sets the mode to be used to acquiring subsequent readings.

Note that the current units will be changed automatically if they are not compatible with the new mode.

Related Commands: MODE_n?, UNITS_n, UNITS_n?

MODE_n?
Acquisition Mode Query

Syntax: MODE_n?

where n is either A or B

Parameters: None

Function:

This query returns the acquisition mode currently being used by the meter.

Returns: <mode>

<mode> is of type <string> in the double quote format.

Related Commands: MODE_n, UNITS_n, UNITS_n?

R? **Read Both Channels Measurement Query**

Syntax: R?

Parameters: None

Function:

This query returns the last valid measurements taken from both channels.

The units of the measurements are the units defined at the time the readings were made. Because it is possible to make this query faster than measurements are actually being taken, the values returned by successive queries may be multiple reports of a single measurements.

To ensure that fresh data is being read the NVDA(new valid data available) bit in the Status Byte can be checked either by issuing a serial poll or by using the *STB? command. A serial poll is recommended if the GPIB interface is being used. The NVDA bit will be 1 if the last valid measurement has not been read. The NVDA bit for a specific channel will be reset to 0 immediately after a measurement is queried from that channel and will stay 0 until a new valid reading is taken. A valid reading is a reading that would return a status of 0 using the RWS? query.

Returns: <number>, <number>
or <number>

<number> is a floating point number in the exponential format.

If two channels are running simultaneously then, both are returned separated by a comma.

Related Commands: *STB?, MODE_n, MODE_n?, R_n?, RWS?, RWS_n?, UNITS_n, UNITS_n?

R_n? **Read Single Channel Measurement Query**

Syntax: R_n?

where n is either A or B

Parameters: None

Function:

This query returns the last valid measurements taken from the specified channel.

The units of the measurements are the units defined at the time the reading was made. Because it is possible to make this query faster than measurements are actually being taken, the values returned by successive queries may be multiple reports of a single measurements. To ensure that fresh data is being read the NVDA(new valid data available) bit in the Status Byte can be checked either by issuing a serial poll or by using the *STB? command. A serial poll is recommended if the GPIB interface is being used. The NVDA bit will be 1 if the last valid measurement has not been read. The NVDA bit for the specified channel will be reset to 0 immediately after a measurement is queried from that channel and will stay 0 until a new valid reading is taken. A valid reading is a reading that would return a status of 0 using the RWS? query.

Returns: <number>

<number> is a floating point number in the exponential format.

Related Commands: *STB?, MODE_n, MODE_n?, R?, RWS?, RWS_n?, UNITS_n, UNITS_n?

RANGE_n

Signal Range Select

Syntax: RANGE_n <range>

where n is either A or B

Parameters:

The parameter <range> is of type <number>. The range of this parameter will depend on the detector and mode being used to acquire data. See the section entitled “Electrical Specifications” at the beginning of this manual for the signal range that corresponds to the current setting of <range>.

Function:

This command is used to select a new manual signal range. This command will disable the auto ranging feature.

Related Commands: AUTO_n, AUTO_n?, RANGE_n?

RANGE_n?

Signal Range Query

Syntax: RANGE_n?

where n is either A or B

Parameters: None

Function:

This query returns a value that represents the current signal range setting being used whether the meter is auto ranging or in manual ranging mode. See Section 2.3.11.

Returns: <range>

<range> is of type <number> and represents the gain level currently being used for either auto or manual ranging. See the section entitled “Electrical Specifications” at the beginning of this manual for the signal range that corresponds to <range>.

Related Commands: AUTO_n, AUTO_n?, RANGE_n

REFSEL_n

Reference Source Select

Syntax: REFSEL_n <source>

Parameters:

<source> is of type <string> in the double quote format. The possible values of <source> are: “STOREF” - Last stored reference measurement, “OTHERCH” - The other channel, “USRREF” - User defined reference value

Function:

This command is used to select a reference source to be used in linear and logarithmic (dB) relative measurements. This command will cause an error if an attempt is made to select the other channel as a reference if it is not available.

Returns: None

Related Commands: REFSEL_n?, STOREF_n, USRREF_n, USRREF_n?

REFSEL_n?

Reference Source Query

Syntax: REFSEL_n?

Parameters: None

Function:

This query returns the current reference source that will be used in linear and logarithmic (dB) relative measurements. Returns:

<source> is of type <string> in the double quote format. The possible values of <source> are:

“STOREF” - Last stored reference measurement

“OTHERCH” - The other channel

“USRREF” - User defined reference value

Related Commands: REFSEL_n, STOREF_n, USRREF_n, USRREF_n?

RESP_n?

Responsivity Query

Syntax: RESP_n?

where n is either A or B

Parameters: None

Function:

This query returns the responsivity value currently being used by the 2835-C in measurement calculation.

Returns: <val>

<val> is of type <number> and represents a floating point number. The units of the responsivity <val> depends upon the type of detector family of the detector in use. See Table 22 for a listing of responsivity units by detector family.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, LAMBDA_n?, USRCAL_n, USRCAL_n?, USRRESP_n, USRRESP_n?

RUN

Start Both Channel Acquisition

Syntax: RUN

Parameters: None

Function:

This command initiates data acquisition on both channels using the currently defined acquisition mode. In single type acquisition modes this command will cause one reading to be acquired. In continuous type acquisition modes this command will initiate the continuous acquisition of readings. The STOP command will terminate acquisition.

Related Commands: STOP, STOP_n, RUN_n

RUN_n **Start Single Channel Acquisition**

Syntax: RUN_n

Parameters: None

Function:

This command initiates data acquisition on the specified channel using the currently defined acquisition mode. In single type acquisition modes this command will cause one reading to be acquired. In continuous type acquisition modes this command will initiate the continuous acquisition of readings. The STOP command will terminate acquisition.

Related Commands: STOP, STOP_n, RUN

RWS? **Read Both Channel Measurements With Status Query**

Syntax: RWS?

Parameters: None

Function:

This query returns the last measurements taken from both channels. The units of these measurements are the units defined at the time the readings were made. Because it is possible to make this query faster than measurements are actually being taken, the values returned by successive queries may be multiple reports of the same measurement.

To ensure that fresh data is being read the NDA(new data available) bit in the Status Byte can be checked either by issuing a serial poll or by using the *STB? command. A serial poll is recommended if the GPIB interface is being used. The NDA bit will be 1 if the last measurement has not been read. The NDA bit for a specific channel will be reset to 0 immediately after a measurement is queried from that channel and will stay 0 until a new reading is taken.

Returns: <status>, <number>

or <status>, <status>, <number>, <number>

<status> is an integer that represents the status of the returned value. <status> will be 0 for ok, 1 for over range, 2 for saturated and 3 for data error, and 4 for ranging.

If more than one channel is available, then the first two numbers represent the status of CHA and CHB respectively.

<number> is a floating point number in the exponential format.

Related Commands: *STB?, MODE_n, MODE_n?, R?, R_n?, RWS_n?

RWS_n?

Read Single Channel Measurement With Status Query

Syntax: RWS_n?

Parameters: None

Function:

This query returns the last measurement taken from the specified channel. The units of this measurement are the units defined at the time the reading was made. Because it is possible to make this query faster than measurements are actually being taken, the values returned by successive queries may be multiple reports of the same measurement.

To ensure that fresh data is being read the NDA(new data available) bit in the Status Byte can be checked either by issuing a serial poll or by using the *STB? command. A serial poll is recommended if the GPIB interface is being used. The NDA bit will be 1 if the last measurement has not been read. The NDA bit for the specified channel will be reset to 0 immediately after a measurement is queried from that channel and will stay 0 until a new reading is taken.

Returns: <status>,<number>

<status> is an integer that represents the status of the returned value. <status> will be 0 for ok, 1 for over range, 2 for saturated and 3 for data error, and 4 for ranging.

<number> is a floating point number in the exponential format.

Related Commands: *STB?, MODE_n, MODE_n?, R?, R_n?, RWS? UNITS_n, UNITS_n?

SFREQ

DC Sample Frequency Select

Syntax: SFREQ <freq>

Parameters:

The parameter <freq> is of type <number>. The range of this number depends on the current DC sample precision (see SPREC). If the precision is 20000 then the range is 0.001 to 25Hz. If the precision is 4096 then the range is 0.001 to 500Hz when two detector calibration modules are plugged into the meter and 1000Hz for one.

Function:

This command determines how often a signal is sampled when in DC continuous mode. Both channels sample data at the same rate when in DC continuous mode.

Related Commands: SFREQ?, SPREC, SPREC?

SFREQ?
DC Sample Frequency Query

Syntax: SFREQ?

Parameters: None

Function:

This query returns the sample rate used for continuous DC acquisition on both channels

Returns: <freq>

<freq> is of type <number> in exponent notation with units of hertz or samples per second.

Related Commands: SFREQ, SPREC, SPREC?

SPREC
DC Sample Precision Select

Syntax: SPREC <prec>

Parameters:

The parameter <prec> is of type <number> with acceptable values of 20000 or 4096.

Function:

The 2835-C has a 20,000 count high precision A/D and a 4,096 count A/D. This command determines which of these A/Ds will be used from both channels to acquire samples during DC single and DC continuous acquisition modes. This command affects the available range of DC sample frequencies.

Note that the 4,096 count A/D is used during all other acquisition modes.

Related Commands: SFREQ, SFREQ?, SPREC?

SPREC?
DC Sample Precision Query

Syntax: SPREC?

Parameters: None

Function:

This query returns a string that indicates whether the 20,000 count A/D or the 4,096 count A/D will be used during DC CONT or DC SNGL acquisition modes on both channels.

Returns: <prec>

<prec> is of type <number>. It will be 20000 if the 20,000 count A/D is selected and 4096 if the 4,096 count A/D is selected.

Related Commands: SFREQ, SFREQ?, SPREC?

STMAX_n?

Statistics Buffer Maximum Value Query

Syntax: STMAX_n?

where n is either A or B

Parameters: None

Function:

This query returns the maximum value in the statistics buffer of the specified channel.

Returns: <max>

<max> is of type <number> in exponent notation.

Related Commands: STMEAN_n?, STMIN_n?, STMXMN_n?, STSDEV_n?, STSIZE_n, STSIZE_n?,

STMEAN_n?

Statistics Buffer Mean Value Query

Syntax: STMEAN_n?

where n is either A or B

Parameters: None

Function:

This query returns the mean or average of all the values in the statistics buffer of the specified channel.

Returns: <mean>

<mean> is of type <number> in exponent notation.

Related Commands: STMAX_n?, STMIN_n?, STSDEV_n?, STMXMN_n?, STSIZE_n, STSIZE_n?

STMIN_n?

Statistics Buffer Minimum Value Query

Syntax: STMIN_n?

where n is either A or B

Parameters: None

Function:

This query returns the minimum value in the statistics buffer of the specified channel.

Returns: <min>

<min> is of type <number> in exponent notation.

Related Commands: STMAX_n?, STMEAN_n?, STSDEV_n?, STMXMN_n?, STSIZE_n, STSIZE_n?

STMXMN_n?

Statistics Buffer Max-Min Query

Syntax: STMXMN_n?

where n is either A or B

Parameters: None

Function:

This query returns the difference between the maximum and minimum readings in the statistics buffer of the specified channel.

Returns: <mxmn>

<mxmn> is of type <number> in exponent notation.

Related Commands: STMAX_n?, STMEAN_n?, STMIN_n?, STMXMN_n, STSIZE_n, STSIZE_n?

STSDEV_n?

Statistics Buffer Standard Deviation Query

Syntax: STSDEV_n?

where n is either A or B

Parameters: None

Function:

This query returns the standard deviation of the readings in the statistics buffer of the specified channel.

Returns: <stddev>

<stddev> is of type <number> in exponent notation.

Related Commands: STMAX_n?, STMEAN_n?, STMIN_n?, STMXMN_n, STSIZE_n, STSIZE_n?

STSIZE_n

Statistics Buffer Size Select

Syntax: STSIZE_n <size>

where n is either A or B

Parameters:

The parameter <size> is of type <number> which rounds to an integer in the range of 0 to 100. The parameter represents the size of the statistics buffer.

Function:

This command sets the size of the buffer used for calculating real-time statistics. The contents of this buffer are updated every time a reading is taken. The statistics buffer is cleared automatically when this command is issued or when the acquisition mode or units are changed. If the buffer size is set to 0 then the statistics are disabled and no statistical information will be maintained.

Related Commands: STMAX_n?, STMEAN_n?, STMIN_n, STSIZE_n?, STSDEV_n?

STSIZE_n?
Statistics Buffer Size Query

Syntax: STSIZE_n?

where n is either A or B

Parameters: None

Function:

This query returns the statistics buffer size.

Returns: <size>

<size> is of type <number> and represents an integer in the range 0 to 100.

Related Commands: STMAX_n?, STMEAN_n?, STMIN_n?, STMXMN_n?, STSDEV_n?, STSIZE_n

STOP
Stop Both Channel Acquisition

Syntax: STOP

Parameters: None

Function:

This command terminates any acquisition of data currently in progress on both channels. If data is not being acquired on a channel then this command has no effect.

Related Commands: RUN, RUN_n, STOP_n

STOP_n
Stop Single Channel Acquisition

Syntax: STOP_n

where n is either A or B

Parameters: None

Function:

This command terminates any acquisition of data currently in progress on the specified channel. If data is not being acquired on that channel the command has no effect.

Related Commands: RUN, RUN_n, STOP

STOREF_n
Store Reference Reading

Syntax: STOREF_n

where n is either A or B

Parameters: None

Function:

This command takes the latest reading and stores it as a reference reading to be used when making relative linear and dB measurements. The units of the reference reading defaults to Watts for Low-Power and High-Power detectors and Joules for Energy detectors.

Related Commands: REFSEL_n, REFSEL_n?, USRREF_n, USRREF_n?, STOREF_n?

STOREF_n? **Store Reference Reading Query**

Syntax: STOREF_n?

where n is either A or B

Parameters: None

Function:

This query returns the latest reading stored as a reference reading. The units of the reference reading defaults to Watts for Low-Power and High-Power detectors and Joules for Energy detectors.

Returns: <number>

<number> is a floating point number in the exponential format.

Related Commands: REFSEL_n, REFSEL_n?, USRREF_n, USRREF_n?, STOREF_n

STOZERO_n **Store Zero Signal**

Syntax: STOZERO_n

where n is either A or B

Parameters: None

Function:

This command takes the latest reading and stores it as a zero signal reference value to be used when zeroing is enabled. The units of the zero reference defaults to Amperes for Low-Power detectors and Volts for High-Power and Energy detectors.

Related Commands: ZERO_n, ZERO_n?, ZEROVAL_n?

TONE **Tone Enable**

Syntax: TONE <state>

Parameters:

The parameter <state> is <number> which can be 0 or 1. If the number is 0 then the tone is disabled. If the number is 1, then tone is enabled.

Function:

This command enables or disables a tone feature that emits a short audible “tick” when a peak or pulse is detected in peak-to-peak or pulse acquisition modes. This command will cause an error if the detector does not support peak-to-peak or pulse acquisition modes.

Related Commands: TONE?

TONE? **Tone Enable Query**

Syntax: TONE?

Parameters: None

Function:

This query returns a value indicating whether the tone feature is not available, enabled or disabled.

Returns: <state>

<state> is a <number> which can be -1, 0 or 1. If -1 then the tone feature is not available for the detector in use. If 0 then the tone feature is disabled. If 1 then the tone feature is enabled.

Related Commands: TONE

TRIGOUT **Trigger Out Mode Select**

Syntax: TRIGOUT <mode>

Parameters:

The parameter <mode> is of type <string>. The valid values for this parameter are listed below.

"CMPLT"	specifies conversion complete mode
"FREQ"	specifies frequency mode
"LVL"	specifies level comparator mode
"INTG"	specifies integration level mode
"OFF"	specifies off mode

Function:

This command determines in which mode the trigger out output on the back panel will operate.

Certain modes require additional parameters to be set.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUT? **Trigger Out Mode Query**

Syntax: TRIGOUT?

Parameters: None

Function:

This query returns the current mode of the trigger out output.

Returns: <mode>

<mode> is of type <string> in the double quote format.

Related Commands: TRIGOUT, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTFREQ

Trigger Out Frequency Select

Syntax: TRIGOUTFREQ <freq>

Parameters:

The parameter <freq> is of type <number>. The range of this number is 0.001 to 1000Hz.

Function:

This command is used to set the frequency at which a pulse is output, on the TTL OUT line, in trigger out frequency mode.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTFREQ?

Trigger Out Frequency Query

Syntax: TRIGOUTFREQ?

Parameters: None

Function:

This query returns the trigger out pulse frequency used during trigger out frequency mode.

Returns: <freq>

<freq> is of type <number> in exponent notation with units of hertz.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTINTG

Trigger Out Integration Level Select

Syntax: TRIGOUTINTG <level>

Parameters:

The parameter <level> is of type <number> which specifies an integration energy level in joules.

Function:

This command is used to set the integration energy level at which the trigger out will become inactive if using the trigger out in integration mode.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTINTG? **Trigger Out Integration Level Query**

Syntax: TRIGOUTINTG?

Parameters: None

Function:

This query returns the trigger out integration energy level at which the trigger out will become inactive if using the trigger out in integration acquisition mode.

Returns: <level>

<level> is of type <number> in exponent notation with units of joules.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTLVL **Trigger Out Level Select**

Syntax: TRIGOUTLVL <level>

Parameters:

The parameter <level> is of type <number>. The units of this parameter are Watts for Low-Power and High-Power detectors and Joules for Energy detectors.

Function:

This command is used to set the level used by the trigger out level comparator mode. When the power or energy level is at or above this level the trigger out will go to its active state.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL?, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTLVL? **Trigger Out Level Query**

Syntax: TRIGOUTLVL?

Parameters: None

Function:

This query returns the trigger out level used by the trigger out level comparator mode.

Returns: <level>

<level> is of type <number> in exponent notation with units of Watts for Low-Power and High-Power detectors and Joules for Energy detectors.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTPOL, TRIGOUTPOL?

TRIGOUTPOL

Trigger Out Polarity Select

Syntax: TRIGOUTPOL <state>

Parameters:

The parameter <state> is a <number> which can be 0 or 1. If the number rounds to 0, then the polarity will be active low. If the number is 1, then the polarity will be active high.

Function:

This command is used to set the polarity of the trigger out output to either active low or active high.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL?

TRIGOUTPOL?

Trigger Out Polarity Query

Syntax: TRIGOUTPOL?

Parameters: None

Function:

This query returns a value that indicates whether the trigger out polarity is active high or active low.

Returns: <state>

<state> is of type <number> and represents the integer 0, if the polarity is active low, or 1, if the polarity is active high.

Related Commands: TRIGOUT, TRIGOUT?, TRIGOUTFREQ, TRIGOUTFREQ?, TRIGOUTINTG, TRIGOUTINTG?, TRIGOUTLVL, TRIGOUTLVL?, TRIGOUTPOL

UNITS_n

Units Select

Syntax: UNITS_n <units>

where n is either A or B

Parameters:

<units> is of type <string>. All possible values of units are listed below:

"V"	specifies volts
"A"	specifies amps
"W"	specifies watts
"W/cm"	specifies watts/cm ²
"dBm"	specifies dBm
"dB"	specifies dB
"REL"	specifies linear ratio
"J"	specifies joules
"J/cm"	specifies joules/cm ²
"ERG"	specifies ergs
"E/cm"	specifies ergs/cm ²

A subset of these units will be valid for a particular detector and acquisition mode.

Function:

Sets the units to be used for subsequent measurements. Once the units have been set, all new readings are given in the new units.

Related Commands: MODE_n, MODE_n?, UNITS_n?

UNITS_n?

Units Query

Syntax: UNITS_n?

where n is either A or B

Parameters: None

Function:

This query returns the units of readings currently being taken.

Returns: <units>

<units> is of type <string> in the double quote format. The possible values of units are: "A", "V", "W", "W/cm", "dBm", "dB", "REL", "J", "J/cm", "ERG", or "E/CM" depending on the detector and acquisition mode.

Related Commands: MODE_n, MODE_n?, UNITS_n

USRCAL_n
User Calibration Enable

Syntax: USRCAL_n <state>

where n is either A or B

Parameters:

The parameter <state> is a <number> which can be 0 or 1. If the number is 0, then user calibration is disabled. If the number is 1, then user calibration is enabled.

Function:

This command enables or disables the use of a user defined calibration point.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, LAMBDA_n?, RESP_n?, USRCAL_n?, USRRESP_n, USRRESP_n?

USRCAL_n?
User Calibration Enable Query

Syntax: USRCAL_n?

where n is either A or B

Parameters: None

Function:

This query returns a value showing whether or not user calibration is enabled or disabled.

Returns: <state>

<state> is of type <number> and represents an integer 0, if user calibration is disabled, or 1, if user calibration is enabled.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, LAMBDA_n?, RESP_n?, USRCAL_n, USRRESP_n, USRRESP_n?

USRREF_n
User Defined Reference Value Select

Syntax: USRREF_n <val>

where n is either A or B

Parameters:

The parameter <val> is of type <number>.

Function:

This command provides a means of directly storing a reference value to be used in linear and logarithmic (dB) relative measurements. The units of this value are watts for Low Power and High Power detectors and joules for Energy detectors.

Related Commands: REFSEL_n, REFSEL_n?, STOREF_n, USRREF_n?

USRREF_n?

User Defined Reference Value Query

Syntax: USRREF_n?

where n is either A or B

Parameters: None

Function:

This query returns the user defined reference value. This value will be in watts for Low-Power and High-Power detectors and joules for Energy detectors.

Returns: <refval>

<refval> is of type <number> in exponential notation.

Related Commands: REFSEL_n, REFSEL_n?, STOREF_n, USRREF_n

USRRESP_n

User Responsivity Factor Select

Syntax: USRRESP_n <val>

where n is either A or B

Parameters:

The parameter <val> is of type <number>. The units of this parameter match the responsivity units of the detector in use.

Function:

This command is used to define a responsivity factor that will be used by the 2835-C in measurement calculation when user calibration is enabled.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, LAMBDA_n?, RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n?

USRRESP_n?

User Responsivity Factor Query

Syntax: USRRESP_n?

where n is either A or B

Parameters: None

Function:

This query returns the user defined responsivity factor that will be used by the 2835-C in measurement calculation when user calibration is enabled.

Returns: <val>

<val> is of type <number> and represents a floating point number. The units of <val> match the responsivity units of detector in use.

Related Commands: ATTN_n, ATTN_n?, LAMBDA_n, LAMBDA_n?, RESP_n?, USRCAL_n, USRCAL_n?, USRRESP_n

ZERO_n **Signal Zeroing Enable**

Syntax: ZERO_n <state>

where n is either A or B

Parameters:

The parameter <state> is a <number> which can be 0 or 1. If the number is 0, then readings are not adjusted by the stored zero reference. If the number is 1, then readings are adjusted by the stored zero reference.

Function:

This command enables or disables the zeroing feature. Zeroing causes the stored zero reference (see STOZERO_n command) to be subtracted from incoming signal readings before the incoming signal is used in measurement calculations.

Related Commands: STOZERO_n, ZERO_n?, ZEROVAL_n

ZERO_n? **Signal Zeroing Enable Query**

Syntax: ZERO_n?

where n is either A or B

Parameters: None

Function:

This query returns a value showing whether or not zeroing is enabled or disabled.

Returns: <state>

<state> is of type <number> and represents the integer 0, if zeroing is enabled, or 1, if zeroing is disabled.

Related Commands: STOZERO_n, ZERO_n, ZEROVAL_n

ZEROVAL_n? **Zero Signal Query**

Syntax: ZEROVAL_n?

where n is either A or B

Parameters: None

Function:

This query returns the current value being used for zeroing.

Returns: <zeroval>

<zeroval> is of type <number> in exponential notation. The units of this value are the numerator of the detector's responsivity, i.e. amps for Low Power detectors and volts for High Power and Energy detectors.

Related Commands: STOZERO_n, ZERO_n, ZERO_n?

Section 6

Maintenance, Test and Troubleshooting

6.1 Maintenance Procedures

In cleaning the body of this instrument, use only a mild soap and water solution on a damp cloth.

CAUTION

Do not use acetone or other organic solvents (other than alcohol) on the Model 2835-C Multi-Function Optical Meter. Organic solvents attack the paint.

6.2 Power Up Self Test

The Model 2835-C executes a system self test after every power up or reset and indicates on the display whether the self test has been passed. If the self test is not passed the display will indicate one of the following error messages:

“CAL MODULE A” “MISSING OR BAD”
“CAL MODULE B” “MISSING OR BAD”
“CAL MODULE A” “CHECKSUM ERROR”
“CAL MODULE B” “CHECKSUM ERROR”
“REGISTER ERR A”
“REGISTER ERR B”
“INTG A/D ERR A”
“INTG A/D ERR B”
“SAA/D ERR A”
“SAA/D ERR B”

If the calibration module messages appear, check to see that the module is properly inserted.

Should any other message appear, call Newport Corporation’s Customer Service Department.

6.3 Troubleshooting Guide

The following troubleshooting guide is intended to isolate and solve problems with the power meter so that, to the greatest extent possible, the return of the power meter/detector system to Newport will be unnecessary. For the problems that cannot be resolved with information in this manual, or for other situations that are not covered in this section, please see Section 7 for details on returning your entire system to Newport for service.

Table 27–Symptom/Fault Troubleshooting Guide

Symptom	Possible Fault/Correction
Blank display.	Power switch OFF. Turn switch ON. Power cord connection is absent. Check power cord connection. Display is blanked. Press DISP key.
Display shows “OL”	Indicates that the signal is too large for the selected signal range. Select a higher RANGE or use an attenuator if one is available.
Display shows “SA”	Indicates that the signal exceeds the detector saturation or damage threshold. Select a detector with higher power or energy handling capability or use an attenuator if one is available.
Display shows “-----”	Indicates that the meter is in the middle of a range change or that the current units don’t match the units of the last reading.
Any of the following messages: “REGISTERERRA” “REGISTERERRB” “INTG A/DERRA” “INTG A/DERRB” “SA A/DERRA” “SA A/DERRB”	Call Newport Corporation’s Customer Service Department and arrange to return the unit for repair.

The display shows:

“CAL MODULE A”
or “CAL MODULE B”
and “MISSING OR BAD”

Make sure the CAL MODULE is seated correctly and reset the meter. Try another CAL MODULE, if one is available, to localize the problem to the first CAL MODULE.

If the first CAL MODULE does not work after proper seating and reset, but the second CAL MODULE works, call Newport Customer Service and arrange for a repair of the first CAL MODULE.

If neither the first or second CAL MODULE works, call Newport Customer Service and arrange for a meter repair. Be sure to also return the first CAL MODULE if you cannot otherwise determine that it is working.

The display shows:

“CAL MODULE A”
“CAL MODULE B”
and “CHECKSUM ERROR”

Call Newport Customer Service and or arrange to return the meter for repair.

The display shows:

“BATTERY FAILURE”

Call Newport Customer Service and arrange to return the meter for battery replacement.

RS-232 communication does
not seem to work

Check the RS-232 cable connection and cable pinouts, Figure 17. Make sure that the device talking to the meter is setup for 8 data bits, no parity, and 1 stop bit. Also check the echo mode state.

No listener error when attempting
GPIB communications

Check the GPIB cable connections and the GPIB address.

Display value does not change.

Press the R/S, R/S A, or R/S B key.

Reading is different than
expected.

See Table 23, Common Measurement Errors. (Section 3.10.6.)

Section 7

Factory Service

7.1 Introduction

This section contains information regarding obtaining factory service for the Model 2835-C Multi-Function Optical Meter. The user should not attempt any maintenance or service of this instrument and/or accessories beyond the procedures given in Section 6: Maintenance, Test and Troubleshooting. Any problems which cannot be resolved using the guidelines listed in Section 6 should be referred to Newport Corporation factory service personnel. Contact Newport Corporation or your Newport representative for assistance.

The Model 2835-C contains no user serviceable parts. Its calibration accuracy is warranted for a period of 1 year. After 1 year, the unit should be returned to Newport Corporation for recalibration and NIST traceability recertification.

7.2 Obtaining Service

To obtain information concerning factory service, contact Newport Corporation or your Newport representative. Please have the following information available:

1. Instrument model number (On front panel)
2. Instrument serial number (On rear panel)
3. Description of the problem.

If the instrument is to be returned to Newport Corporation, you will be given a Return Number, which you should reference in your shipping documents.

Please fill out the service form, located on the following page, and have the information ready when contacting Newport Corporation. Return the completed service form with the instrument.

Service Form

Newport Corporation
U.S.A. Office: 714/863-3144
FAX: 714/253-1800

Name _____ RETURN AUTHORIZATION # _____
Company _____ (Please obtain prior to return of item)
Address _____
Country _____ Date _____
P.O. Number _____ Phone Number _____

Item(s) Being Returned:

Model # _____ Serial # _____

Description _____

Reason for return of goods (please list any specific problems) _____

List all control settings and describe problem _____

(Attach additional sheets as necessary).

Show a block diagram of your measurement system including all instruments connected (whether power is turned on or not). Describe signal source. If source is a laser, describe output mode, peak power, pulse width, repetition rate and energy density.

Where is the Measurement Being Performed?

(factory, controlled laboratory, out-of-doors, etc.) _____

What power line voltage is used? _____ Variation? _____

Frequency? _____ Ambient Temperature? _____

Variation? _____ °F. Rel. Humidity? _____ Other? _____

Any additional information. (If special modifications have been made by the user, please describe below).



Appendix A

Syntax and Definitions

A.1 Definition of <string>

The IEEE 488.2 standard defines two types of string data. These strings are either single or double quoted. For convenience, the 2835-C also recognizes an unquoted string with certain restrictions as detailed below. Any of these forms may be used where a <string> parameter is required.

1. <string>, using double quotes. **"this is a string"**
2. <string>, using single quotes. **'this is a string'**
3. <string>, using no quotes. **thisisastring**

A description of each type of <string> follows:

1. <string> defined using double quotes.

A double quote indicates that a string follows, and the string is terminated by another double quote. A double quote may be embedded within the string by using two double quotes together:

Example: "this string contains a "" double quote"

All characters within the two outer double quotes are considered part of the string. It is an error if the string does not terminate with a double quote. The string cannot contain the <CR>(ascii decimal 13), <LF>(ascii decimal 10), or End or Identify characters.

2. <string> defined using single quotes.

This form is similar to double quoted string. A single quote indicates that a string follows, and the string is terminated by another single quote. A single quote may be embedded within the string by using two single quotes together:

Example: 'this string contains a "' single quote'

All characters within the two outer single quotes are considered part of the string. It is an error if the string does not terminate with a single quote. The string cannot contain the <CR>(ascii decimal 13), <LF>(ascii decimal 10), or End or Identify characters.

3. *<string>* defined using no quotes.

All strings using this format must start with an alphabetic character (A through Z, a through z). All other characters must be either alphabetic, digit (0 through 9) or the '_' character. Any other character will delimit the string.

Some examples are shown below:

Sent:	this is a string	
Interpreted:	this	(1st string)
	is	(2nd string)
	a	(3rd string)
	string	(4th string)
Sent:	this,isastring	
Interpreted:	this	(1st string)
	,	(separator character)
	isastring	(2nd string)
Sent:	w/cm	
Interpreted:	w	(1st string)
	ERROR	(unrecognized character)
	cm	(2nd string)

A.2 Definition of *<number>*

The IEEE 488.2 standard defines four different types of numeric data. The 2835-C recognizes all four types as *<number>*, thus any format may be used.

1. *<number>* defined as floating point.
2. *<number>* defined as binary.
3. *<number>* defined as octal.
4. *<number>* defined as hexadecimal.

Where necessary, integers are converted to floating point numbers. In all cases, a number is terminated by any of the below characters:

, ; <NL> <EOI> <SPACE>

Any non-valid characters detected in any number received are considered an error in format, and an error condition will be generated in the system. There are no differences between the 2835-C and IEEE-488.2 standard for number definition.

A description of each type of *<number>* follows:

1. *<number>* defined as floating point.

Any of the following characters, as the first character of an ASCII sequence, indicates that a number is being defined:

+ - . 0 1 2 3 4 5 6 7 8 9

A floating point number is defined as follows:

1. Optional + - sign. This defines the sign of the number. If missing, positive is assumed.
2. Optional 0-9 digits. These digits define the integer portion of the mantissa.
3. Optional . decimal point. This defines the end of the integer portion of the mantissa, and indicates that the fractional portion of the mantissa follows.
4. Optional 0-9 digits. These digits define the fractional portion of the mantissa.
5. Optional exponent indicator, an ASCII 'E' or 'e', followed by a '+' or '-' (optional), followed by decimal digits.

Examples:

the numbers below all represent the value "1.2"

1.2

1.2e0

+01.2E+00000

120E-2

.12e1

the numbers below all represent the value "-1.2"

-1.2

-1.2e+00

-0001.2e+0

-120e-2

.12E1

2. <number> defined as binary.

The 2835-C recognizes unsigned binary numbers in the range of 0 to 65535, decimal, or 0 to 1111111111111111 binary. Binary numbers are represented using only the digits 0 and 1. A binary number has the following format:

#B<binary>

Where

#B = mandatory binary number header

<binary> = binary digits (0's or 1's)

Example:

All numbers below represent the decimal value 129.

#B10000001

#b010000001

#b10000001

3. *<number>* defined as octal.

The 2835-C recognizes unsigned octal numbers in the range 0 to 65535 decimal, or 0 to 177777 octal. Octal numbers are represented using digits from 0 to 7. An octal number has the following format:

#Q<octal>

Where

#Q = mandatory octal number header

<octal> = octal digits (0 to 7)

Example:

All numbers below represent the decimal value 129.

#Q201

#q0201

#q201

4. *<number>* defined as hexadecimal.

The 2835-C recognizes unsigned hexadecimal numbers in the range 0 to 65535 decimal, or 0 to FFFF hexadecimal). Hexadecimal numbers are represented using the digits 0-9 and the characters A-F. A hexadecimal number has the following format:

#H<hexadecimal>

Where

#H = mandatory octal number header

<hexadecimal> = hexadecimal digits (0-9 and A-F)

Example:

All numbers below represent the decimal value 127.

#H7f

#H007F

#h7f

Appendix B

Error Messages

The IEEE 488.2 standard defines certain bits in the status registers as error condition flags. When an error occurs, one of the error bits is set in the status registers. The bit enable masks and the service request enable allow the 2835-C to alert the remote controller that an error has occurred. The standard allows error numbers that range from -100 to -499.

When using the RS-232C port the RS-232 Echo Mode controls when errors are returned. When the Echo Mode is enabled the errors are returned immediately. When the Echo Mode is disabled the errors are not returned immediately and the *ERR? command must be used to retrieve the errors.

B.1 Command Errors

Command Errors are associated with the conversion of the data received into the commands and their parameters (parsing). Incorrect syntax, incorrect parameters, and improper command format will generate these errors. Any command error will cause the Command Error bit (bit 5) in the Standard Event Status Byte to be set.

-102, "Syntax error"

Occurs when an error in command structure or parameter type is detected. Some typical causes are:

- Using a number as a command mnemonic.
- Using the wrong parameter type.
- Using ASCII characters outside of a string constant that are not defined by the command language syntax.
- Missing or too many parameters.

The above list is not exhaustive but does give the basic idea of what to look for.

-110, "Command header error"

This error is generated when the system parser detects an invalid character in a command header.

-113, "Undefined Header"

This error is generated when the 2835-C does not recognize a command header.

-121, “Invalid character in number”

This error indicates that a numeric parameter contains an invalid character or a character in the wrong place. The only characters allowed in a number are:

0-9 e E . + -

When using the binary number format, the only characters allowed are:

0 1

When using the octal number format, the only characters allowed are:

0 1 2 3 4 5 6 7

When using the hexadecimal number format, the only characters allowed are:

0-9 A-F a-f

-150, “String data error”

This error is generated when the system parser detects an error in string type data.

-151, “Invalid string data”

Usually caused by an error in string termination. The IEEE 488.2 standard defines all quoted strings as having a closing quote. If the closing quote is missing, usually due to a <NL> character being embedded in the string, this error will be generated.

-191, “Out of string space”

Generated when the system parser runs out of space for commands and parameters. It may occur if commands are not terminated correctly.

-192, “Out of token space”

Occurs when the system parser detects more command elements (headers and parameters) than it can handle. It may also occur if commands are not terminated correctly.

B.2 Execution Errors

Execution Errors are associated with the interpretation of the converted commands and parameters received. Incorrect parameter values and numerical range errors are types of execution errors. Any execution error will cause the Execution Error bit (bit 4) in the Standard Event Status Byte to be set.

-220, “Parameter error”

This is a general parameter error. If this error occurs the parameter may be invalid or inappropriate.

-222, “Data out of range”

This error will occur if a parameter is out of a valid range or not in the set of valid parameters for a given command.

-286, “Configuration not compatible”

This error will occur if an attempt is made to recall a system configuration that is not compatible with the detector currently attached to the meter.

-287, “Configuration not available”

This error will occur if an attempt is made to recall a system configuration that has not been saved to and is therefore not available.

-288, “Command not available”

This error will occur if an attempt is made to use a command that is not available for the detector in use.

B.3 Device Errors

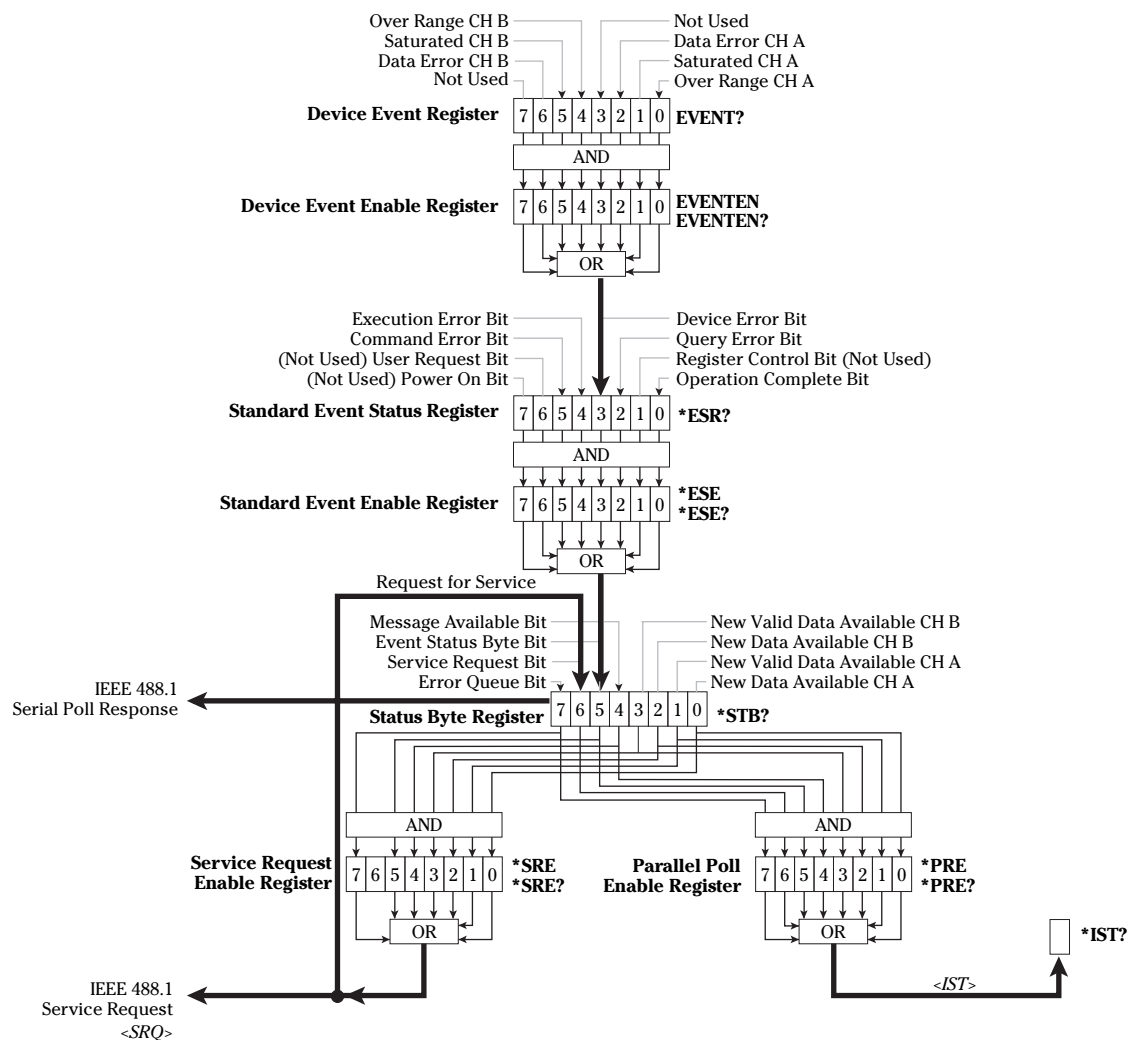
Device Errors are associated with some system condition that affects the operation of the meter. Errors associated with data reading will set the appropriate bit but will not generate an error message to avoid jamming the error queue or the interface.

Appendix C

Status Reporting System

On this page is a graphical representation of the status reporting system for the GPIB and RS-232 port communications. Following is a detailed description of each register and how the bits are set and reset. The interactions between registers is discussed as well as the generation of a GPIB Service Requests.

C.1 Status Reporting System Flowchart



C.2 Device Event Register

The Device Event Register is used to record device errors associated with some system condition that affects the operation of the meter. When a bit is set (has a value of one) then the condition is true. The bit remains set until cleared by the EVENT? query or *CLS command.

The Device Event Enable Register is used to define the conditions that will set the Device Error bit in the Standard Event Status register. If a bit is set in the Device Event Register and its corresponding bit is set in the Device Event Enable Register, then the Device Error bit in the Standard Event register will be set.

The Device Event Enable Register is configured by using the EVENTEN command. The Device Event Register is described below. The register is bit mapped, with each bit representing the described condition. The bits are listed most significant bit first.

Device Event Register

bit 7 Not Currently Used.

bit 6: Data Error Channel B

A one in this bit indicates that a data error occurred on channel B during reading calculations. These errors will include such conditions as divide by zero or taking the log of a negative number.

bit 5: Saturated Channel B

Many detectors contain information in their CAL MODULE that indicates when the detector saturates or is otherwise over driven. A one in this bit indicates that a reading on channel B was taken at or above this saturation level.

bit 4: Overrange Error Channel B

A one in this bit indicates that an overrange condition on channel B occurs when taking a reading.

bit 3: Not Currently Used

bit 2: Data Error Channel A

A one in this bit indicates that a data error occurred on channel A during reading calculations. These errors will include such conditions as divide by zero or taking the log of a negative number.

bit 1: Saturated Channel A

Many detectors contain information in their CAL MODULE that indicates when the detector saturates or is otherwise over driven. A one in this bit indicates that a reading was taken on channel A at or above this saturation level.

bit 0: Overrange Error Channel A

A one in this bit indicates that an overrange condition on channel A occurs when taking a reading.

C.3 Standard Event Status Register

The Standard Event Status Register is used to record general system event conditions for the status reporting system. The register is bit mapped, meaning that each condition is represented by a bit. When a bit is set (has a value of 1), then the condition is true. The bit remains set until cleared by the *ESR? query or the *CLS command.

The Standard Event Enable Register is used to define the conditions that will set the Event Status Byte bit (bit 5) in the Status Byte. If a bit is set in the Standard Event Status register and its corresponding bit is set in the Standard Event Enable Register, then the Event Status Byte bit (bit 5) in the Standard Event Register will be set. The Standard Event Enable Register is configured by using the *ESE common command.

Each of the bits in the Standard Event Status Register is described below.

Standard Event Status Register

bit 7: Power On Not used by the 2835-C.

bit 6: User Request Not used by the 2835-C.

bit 5: Command Error

A 1 in this bit indicates that the 2835-C has received a remote command that generated a command error.

bit 4: Execution Error

A 1 in this bit indicates that the 2835-C has received a remote command that generated an execution error.

bit 3: Device Error

A 1 in this bit indicates that an unmasked device error has occurred.

bit 2: Query Error

A 1 in this bit indicates that a query error has occurred.

bit 1: Request Control. This bit is always 0.

bit 0: Operation Complete

This bit is controlled by the *OPC command. If the *OPC command is in effect, then this bit will be set to 1 when all pending operations have completed. To operate correctly, this bit should be cleared by the *CLS command or *ESR? query before the *OPC command is issued again.

C.4 Status Byte

The Status Byte register is used to record a summary of current system conditions for the status reporting system. It is returned to the controller when a serial poll of the 2835-C is conducted or when the *STB? query is issued. The register is bit mapped, meaning that each condition is represented by a bit. When a bit is set, or has a value of 1, then the condition is true. The bits are cleared based on the conditions described for each bit.

The Service Request Enable register is used to define the conditions that will generate a IEEE 488.1 <SRQ>. When an event occurs that causes a bit to be set in the Status Byte register and its corresponding bit is set in the Service Request Enable register, then a <SRQ> will be generated.

The Service Request Enable Register is configured by using the *SRE common command.

The Parallel Poll Enable Register is used with the Status Byte to generate the <IST> message. If any bit is set in the Status Byte and its corresponding bit is set in the Parallel Poll Enable Register, then the <IST> message is set true (a value of one). Otherwise the <IST> message is set false (a value of zero).

When a parallel poll is conducted with the 2835-C configured to respond to it, the <IST> message is compared to the <S BIT> (sense bit). If they are the same then the configured data line will be driven true in response to the parallel poll.

Both the <S BIT> and the data line driven during a parallel poll can be selected by the IEEE 488.1 parallel poll remote configuration command.

The Parallel Poll Enable Register is configured by using the *PRE command. The Status Byte Register is described below.

Status Byte

bit 7: Error Queue

A one in this bit indicates that the error queue is not empty (see the *ERR? query). The *CLS command will empty the error queue and, as a result, this bit.

bit 6: Service Request/Master Summary Status

When the status byte is read by means of a serial poll, this bit is set when the 2835-C is requesting service.

When the status byte is read by means of the *STB? query, this bit will be set if any bit in the status byte is set and its corresponding bit is set in the Service Request Enable Register.

bit 5: Event Status Byte

The Event Status Byte bit is set when a bit in the Standard Event Status register is set and its corresponding bit in the Standard Event Enable register is set. The *CLS command or *ESR? query will clear the Standard Event Status register and, as a result, this bit.

bit 4: Message Available

The Message Available MAV bit becomes set when any message is ready to be transmitted over the GPIB interface (not the RS-232 interface). Once the message is sent, the MAV bit is cleared. A GPIB device clear command will clear the output queue and, as a result, this bit.

NOTE:

After a query is issued over the GPIB interface this bit should be checked by means of a serial poll before attempting to read the query response.

bit 3: New Valid Data Available

The New Valid Data Available Channel B (NVDA) bit becomes set when a new reading has been taken on channel B that is not overrange, did not saturate the detector, did not cause a data error and was not taken while ranging. It is cleared by the R?, R_A?, RWS? or RWS_A? queries.

bit 2: New Data Available

The New Data Available Channel B (NDA) bit becomes set when a new reading has been taken on channel B. It is cleared by the R?, R_A?, RWS? or RWS_A? queries.

bit 1: New Valid Data Available

The New Valid Data Available Channel A (NVDA) bit becomes set when a new reading has been taken on channel A that is not overrange, did not saturate the detector, did not cause a data error and was not taken while ranging. It is cleared by the R?, R_A?, RWS? or RWS_A? queries.

bit 0: New Data Available

The New Data Available Channel A (NDA) bit becomes set when a new reading has been taken on channel A. It is cleared by the R?, R_A?, RWS? or RWS_A? queries.

Appendix D

Sample Programs

D.1 Example Program RS-232C Communication

```
10      '***** Program Header*****
20      'NEWPORT CORPORATION
30      '2835-C to RS232 Communication Program - an example program
40      '
50      'This program is designed to show you how to write a simple
60      'program that will write commands and read query responses to and
70      'from the 2835-C Multi-Function Optical Meter via the RS-232 port on the
80      'Rear Panel and the RS-232 port on an IBM PC/AT or compatible. The
90      'Program was written in MICROSOFT GWBASIC on an IBM AT compatible.
100     '
110     'Written By: Darwin D. Smith
120     '   Date: April 28, 1993
130     '***** End of Header *****
1000    'Beginning of program
1010        'Open COM port with the following specifications:
1020        'COM port 1, 9600   baudrate, no parity, 8 data bits & 1 stop bit
1030        OPEN "COM2:9600,N,8,1" FOR RANDOM AS #1
1040        GOSUB 2000                'Draw header on the screen
1050        GOSUB 3000                'Process user input
1060        CLOSE #1                 'Close the COM file
1070    END 'End of program
1080    '
2000    'Main.Screen: Draw the main screen.
2010        CLS
2020        LOCATE 1, 20: PRINT "NEWPORT CORPORATION"
2030        LOCATE 2, 20: PRINT "2835-C to RS-232 Communication Program"
2040        LOCATE 3, 20: PRINT "          q or Q to Quit"
2050    RETURN
2060    '
3000    'Enter.User.Commands: Get and interpret the user's commands.
3010        WHILE (1) 'Get and process user input until Q or q is input.
3020            RS232OUT$ = ""                'Clear RS232out$ string
3030            LINE INPUT RS232OUT$          'Get the user input
3040            IF RS232OUT$ = "Q" OR RS232OUT$ = "q" THEN RETURN ELSE GOSUB 4000
3050            'Assume strings ending with "?" are queries
3060            IF INSTR(RS232OUT$, "?") = 0    THEN GOTO 3090
3070                GOSUB 5000
3080            'END IF
3090        WEND
3100    RETURN
3110    '
4000    'Write.RS232.String: Write the string RS232OUT$ to the RS232 port
4010        PRINT #1, RS232OUT$ 'PRINT # appends <CR><LF> to the string
4020    RETURN
4030    '
```

```

5000 'Read.RS232.String: Read a string from the 2835-C RS232 port.
5010 'Characters are read one at a time until a <LF> character is
5020 'read or 8 seconds elapse between consecutive character reads.
5030 'At the end of the routine the string read is in RS232IN$.
5040     BUFFER$ = CHR$(0)           'Initialize buffer$ to NULL character
5050     RS232IN$ = ""
5060     TIMEOUT.ERROR% = 0           'Initiate to no timeout error
5070     ON TIMER(8) GOSUB 6000       'Set timer for a 1 second time out
5080     TIMER ON                     'Turn on timer
5090 'While port doesn't timeout and string delimiter <LF> not
5100 'read, continue trying to read input on COM port.
5110     WHILE TIMEOUT.ERROR% = 0     AND BUFFER$ <> CHR$(10)
5120         IF LOC(1) = 0 THEN GOTO 5170
5130         TIMER OFF
5140         BUFFER$ = INPUT$(1, #1)
5150         RS232IN$ = RS232IN$ + BUFFER$
5160         TIMER ON
5170     '
5180     WEND
5190     TIMER OFF
5200     IF TIMEOUT.ERROR% = 0         THEN GOTO 5230
5210     PRINT "Timed out when reading RS-232 port."
5220     TIMEOUT.ERROR% = 0
5230 '
5240     PRINT RS232IN$;
5250 '
5260 RETURN
5270 '
6000 'timeout: Set TIMEOUT.ERROR% flag. Called if t seconds, as
6010 'defined by ON TIMER(t), in the Read.RS232.String, has elapsed
6020 'between TIMER ON AND TIMER OFF.
6030     TIMEOUT.ERROR% = 1
6040 RETURN
10000 END 'End of Program Listing

```

D.2 Example Program IEEE-488 Communication

```
***** Program Header *****
`NEWPORT CORPORATION
`2835-C to IEEE-488 Communication Program - an example program
`
`The following program is designed to be an example of how to write a simple
`program that will write commands and read query responses to and from
`the 2835-C Multi-Function Optical Meter via the IEEE-488 port.
`
`This program was written on an IBM AT compatible using QuickBASIC 4.5
` and software routines included with the GPIB board.
`The GPIB board used was:   National Instruments GPIB-PC2A board
`                           Newport Corporation pn LA-PC-488-2A-5
`The interface cable was:   Newport Corporation pn LA-CABLE-2M488
`
`NOTE: Before running this program follow National Instrument's instructions
`       for hardware and software installation. This program uses the GPIB
`       configuration device named DEV5 which uses GPIB address 5 by default.
`       All routines that begin with the letters "IB" are defined in the
`       National Instruments library.
`
`Written By: Darwin D. Smith
`       Date: April 28, 1993
***** End of Header *****
`Beginning of program
`$INCLUDE: 'E:\b45\QBDECL.BAS'      'Use your own path here.
`Beginning of program
    IEEEout$ = "DEV5"      'Use this name or assign a new one
    CALL IBFIND(IEEEout$, device.number%) 'Setup the device.number% variable
                                   'used in GPIB-PC routine calls.

    IF device.number% < 0 THEN
        PRINT "Unable to find "; IEEEout$; " device."
        STOP
    END IF
        GOSUB Main.Screen          'Draw header on the screen
        GOSUB Enter.User.Commands  'Process user input
END 'End of program

`Main.Screen: Draw the main screen.
Main.Screen:
    CLS
    LOCATE 1, 22: PRINT "NEWPORT CORPORATION"
    LOCATE 2, 19: PRINT "2835-C to IEEE-488 Communication Program"
    LOCATE 3, 19: PRINT "          Q or q to Quit"
RETURN

`Enter.User.Commands: Get and interpret the user's commands.
Enter.User.Commands:
    DO 'Get and process user input until Q, or QUIT input.
        IEEEout$ = ""      'Clear IEEEout$ string
        LINE INPUT IEEEout$ 'Get the user input
        IEEEout$ = UCASE$(IEEEout$) 'Convert input to upper case
    SELECT CASE IEEEout$
        CASE "Q", "QUIT"      'Exit on Q
            RETURN
        CASE "IBCLR"          'Allow a device clear
            CALL IBCLR(device.number%)
        CASE "IBRSP"          'Allow a serial poll
            CALL IBRSP(device.number%, poll%)
            PRINT poll%
```

```

CASE ELSE
    GOSUB Write.IEEE.String      'Write user input to IEEE port
    IF INSTR(IEEEout$, "?") <> 0 THEN 'If command was a query
        GOSUB Read.IEEE.String   ' Read input data on IEEE port
    END IF
END SELECT
LOOP WHILE 1 = 1
RETURN

'Write.IEEE.String: Write the string IEEEout$ out to the IEEE port.
Write.IEEE.String:
    IEEEout$ = IEEEout$ + CHR$(10)      'Append a <LF> to IEEEout$
    CALL IBWRT(device.number%, IEEEout$) 'Write IEEEout$ to IEEE-488 port
RETURN

'Read.IEEE.String: Read a string from the 2835-C IEEE port. The string
' read from the IEEE-488 port is printed on the screen. If the MAV bit
' in the status byte doesn't go high after 8 seconds then this routine
' will timeout. Otherwise the query response is in IEEEin$
Read.IEEE.String:
    timeout.error% = 0                'Initiate to no timeout error
    poll% = 0
    IEEEin$ = SPACE$(255)             'Initiate IEEEin$ to 255 spaces
    ON TIMER(8) GOSUB timeout

    'wait for MAV bit in the status byte to be set or a timeout to occur
    TIMER ON
    WHILE ((poll% AND &H10) = 0) AND (timeout.error% = 0)
        CALL IBRSP(device.number%, poll%)
    WEND
    TIMER OFF

    IF (timeout.error% = 1) THEN
        PRINT "IEEE-488 port timed out when trying to read input."
    ELSE
        CALL IBRD(device.number%, IEEEin$) 'Read IEEE-488 port
        IEEEin$ = RTRIM$(IEEEin$)         'Trim trailing spaces
        PRINT IEEEin$;                    'Print the query response
    END IF
RETURN

'timeout: Set timeout.error% flag. This routine is called if t seconds,
'as defined by the ON TIMER(t) function call, has elapsed between TIMER ON
'and TIMER OFF.
timeout:
    timeout.error% = 1
RETURN

END 'End of Program Listing

```



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